

CUBIQO — Complete Strategic Report (Part 2 of 2)

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Part 2 covers: Appendix D Topics 5-15 · Appendix E Topics 1-10 · Section 11 (Flagship Spec Gap Analysis with 33-point scorecard)

Part 1 link: [CUBIQO_COMPLETE_REPORT_PART1.pdf](#) on the same branch — Executive Summary, 20 TFRs, Sections 1-8, Appendices A-D Topics 1-4

Topic 5 — Emergent Coding Platform Status

What Actually Exists in Code

Component	File	Status
Projects API (create/list)	/api/emergent/projects/	✓ Full CRUD with auth + validation
Workspaces API	/api/emergent/workspaces/	✓ Working
Files API	/api/emergent/files/	✓ Working
Terminal API	/api/emergent/terminal/	✓ Working
Secrets management	/api/emergent/secrets/	✓ Full CRUD + rotate
Deploy API	/api/emergent/deploy/	✓ Vercel deploy wired
Analytics API	/api/emergent/analytics/	✓ Working
Audit log API	/api/emergent/audit/	✓ Working
Org management	/api/emergent/orgs/	✓ With RBAC
Monaco Editor	LiveCoderPane.tsx	✓ Working — TypeScript/JS/HTML/CSS
Sandbox preview	LiveCoderPane.tsx	✓ CSP-sandboxed iframe
Security RBAC	emergent/security/rbac.ts	✓ Role-based access control
Audit logger	emergent/security/audit-logger.ts	✓ Working
Stacks supported	Schema	TypeScript, JavaScript, Python, Next.js, React, Vue, Svelte, Vanilla

Honest Assessment vs Replit / Vercel v0

WHAT EMERGENT DOES WELL:

- ✓ RBAC with organisations and permission levels
- ✓ Secrets management with rotation

- ✓ Audit logging on all operations
- ✓ Live Monaco editor with syntax highlighting
- ✓ Sandboxed preview (CSP-strict)
- ✓ Vercel deployment integration
- ✓ Project + workspace management

WHAT IS MISSING FOR PROFESSIONAL GRADE:

- ✗ No collaborative editing (multiple cursors / real-time sync)
- ✗ No git integration in the UI (commits, branches, PRs shown in editor)
- ✗ No package manager UI (npm install within the platform)
- ✗ No terminal output streaming (real-time build logs)
- ✗ No debugging tools (breakpoints, step-through)
- ✗ No template library (start from Next.js starter, not blank)
- ✗ No AI inline code completion (Copilot-style)
- ✗ No multi-file navigation sidebar visible
- ✗ Preview only handles vanilla JS – not full Next.js / React rendering
- ✗ No mobile preview mode

What to Build to Make It Professional Grade

Sprint 1 (2-3 weeks) – Core editor UX:

- File sidebar: tree view, create/rename/delete files
- Terminal streaming: SSE-based real-time output (build logs, npm output)
- Template library: 5 starters (Next.js blank, React app, Landing page, API, Blog)
- AI code completion: tab-triggered inline Haiku/Deepseek completions

Sprint 2 (3-4 weeks) – Git integration:

- Connect to GitHub OAuth (already partially implemented in admin/connections)
- Show commits timeline in sidebar
- One-click "open PR" from Emergent to GitHub
- Auto-commit every save (like Replit)

Sprint 3 (2-3 weeks) – Full preview + deploy:

- Next.js preview: spin up a lightweight Next.js process per workspace
- Custom domain assignment for each deployed project (via Vercel API)
- Environment variable UI mapped to Vercel project env vars
- One-click deploy button → shows deploy status in real time

Topic 6 — In-App Agents: Capabilities, Access & Gaps

All 7 Agents (Confirmed in `src/lib/engine/router.ts`)

ID	Name	Trigger Keywords	Capabilities	Access Path
A1	Henry (Orchestrator)	Fallback / coordination	Multi-agent task orchestration, session memory, tool delegation	Default agent — active at all times
A2	Developer	code, dev, build, fix, bug, deploy, implement, refactor, debug	Code generation, file operations, terminal commands, code review	Say "write code" / "build" in chat
A3	Writer	write, document, content, blog, copy, article, draft	Long-form writing, SEO copywriting, email drafts, README	Say "write" / "draft" in chat
A4	Tester			

ID	Name	Trigger Keywords	Capabilities	Access Path
		test, qa, quality, verify, validation, regression, coverage	Test writing, bug reports, validation scripts	Say "test" / "QA" in chat
A5	Marketing	market, social, campaign, growth, promote, SEO, audience	Content calendars, social copy, campaign briefs, SEO articles	Say "marketing" / "campaign" in chat
A6	Animator	animate, visual, 3D, design, motion, graphic, render	Animation scripts, 3D prompts, visual concepts, design briefs	Say "animate" / "design" in chat
A7	Business	outreach, customer, sales, email, lead, CRM, contact	Sales emails, outreach templates, CRM data, business intelligence	Say "outreach" / "sales" in chat

Agent Capabilities in Detail

ALL AGENTS SHARE:

- ✓ Persistent session memory (conversation history stored)
- ✓ Tool registry access (filtered by user permissions)
- ✓ Auto-compaction when context window fills (saves tokens)
- ✓ SOUL.md personality file (agents/A1/SOUL.md etc.)
- ✓ Workspace isolation (each agent has its own file workspace)
- ✓ Multi-concurrent task support (A1: up to 2 parallel tasks)
- ✓ Spawn capability: A1 can spawn A2-A7 for subtasks
- ✓ Session compaction: auto-summarises old history

TOOLS AVAILABLE TO AGENTS (from tool registry):

- ✓ File read/write operations
- ✓ Code execution (sandboxed)
- ✓ Browser automation (Playwright via /api/browser/)
- ✓ Web search (via LLM router with search mode)
- ✓ Email send (via Resend integration)
- ✓ Calendar (if user connects Google Calendar)
- ✓ GitHub (if user connects GitHub)
- ✓ Slack, Discord, Telegram (if user connects)

How Users Access Agents

Current access paths:

1. /agents page – list all agents, start new session
2. Chat: type a message containing trigger keywords → routes automatically
3. /agent-portal – agent management UI
4. Voice: speak trigger keywords → voice routes to correct agent

What needs to be built:

- Agent picker UI in chat: @henry, @dev, @writer inline mentions
- Agent card gallery: visual showcase of each agent's specialty
- Agent output preview: show A3's drafts in a document editor UI
- Agent workspace browser: see A2's files without leaving chat
- Agent collaboration: A2 writes code, A4 auto-tests it, A3 documents it (multi-agent pipeline initiated from single user request)

Topic 7 — Duo Mode & Companion Mode

What Exists in Code

DUO MODE (confirmed):

File: src/components/chat/DuoModeToggle.tsx – UI component exists
Feature flag: duo_mode in feature-gate-simple.ts
Access: Founders only (FOUNDER_ACCESS.duo_mode = true, USER_ACCESS.duo_mode = false)
Description: "Proactive AI interjections – AI interjects with health/safety/tone advice"
UI: Toggle button in chat header with purple glow animation
Backend: The autopilot profile extraction runs in background (/api/autopilot/tasks) while AI companion converses

COMPANION MODE (Sidekick):

Feature flag: sidekick_mode – enabled for founders
File: founderspass/dashboard page shows "Sidekick Companion" as feature
Description: "AI companion mode"
Backend: No dedicated sidekick route found – appears to be a variant of chat mode

COPE MODE:

Feature flag: cope_mode – enabled for founders
Description: Not documented in code, but exists as a feature gate
Likely: Emotional support / mental health focused conversation mode

VOICE MODE:

Feature flag: voice_mode – separate toggle
Working: voice_mode is the ElevenLabs TTS + Whisper STT pipeline

What Needs to Be Configured / Built

Duo Mode – what the proactive interjections should include:

- Health: "You've been talking for 45 minutes – take a breath"
- Safety: Crisis pattern detected (already in PolicyRouter) → proactive check-in
- Tone coaching: "That email sounds aggressive – want me to soften it?"
- Focus: "You've shifted topics 5 times – want to focus on one thing?"
- Finance: "You mentioned money stress 3 times this week – want to journal?"

Companion Mode – what makes it different from standard chat:

- Persistent persona: remembers user's name, preferred topics, past decisions
- Proactive memory: "Last week you said you wanted to learn Python – any progress?"
- Emotional check-in: opens sessions with "How's today going?"
- Voice-default: companion mode should default to voice, not text
- Daily greeting message: pushed at user's typical start time

What to build:

- Duo Mode: connect autopilot tasks to proactive interjection API route
 - Background task checks for health/focus/tone signals every N turns
 - Injects interjection into conversation as assistant message
- Companion Mode: separate system prompt that emphasises continuity + memory
- User settings panel: "Companion personality" slider (professional ↔ personal)

Topic 8 — Job Hunt: Ideal User Flow vs Current Reality

Ideal Complete User Flow

STEP 1 – INTAKE:

User lands on /job-hunt → "Set up your profile"

- Upload resume (PDF/DOCX) ✓ exists at /api/job-hunt/resume
- Enter LinkedIn URL ✓ field exists in JobHuntProfile type
- Select: target roles, locations, work type, salary range ✓ all in schema
- Answer questionnaire (common application questions: "Are you authorized to work?")
✓ /api/job-hunt/questions exists

STEP 2 – AUTOMATED SEARCH:

Cubiqo browser agent goes to:

- LinkedIn Jobs ✓ (Playwright browser + credentials)
- Indeed, Glassdoor, ZipRecruiter ✓ (platforms in JobPlatform type)
- Company career pages (custom browser sessions) ✓ via browser-service.ts

Filters jobs by profile preferences

Scores each job for match (AI scoring against resume)

Returns ranked list to user

STEP 3 – USER REVIEW:

User sees feed of matched jobs with match score

- "Apply automatically" checkbox per job
- "Customise cover letter" option
- "Add to watchlist" (manual review later)

STEP 4 – AUTOMATED APPLICATION:

For each approved job:

- Navigate to job URL via Playwright ✓
- Fill application form fields from profile ✓ (credentials encrypted in DB)
- Upload resume from profile ✓
- Use stored questionnaire answers for common fields ✓
- Submit → store application in job_applications table ✓

STEP 5 – TRACKING & ALERTS:

User gets daily email report: applications submitted, responses received ✓

Interview detected → alert sent ✓ (interview_alert report type exists)

Status tracking: pending → applied → screening → interview → offer/rejected ✓

STEP 6 – INTERVIEW PREP:

Not yet implemented – should be:

- AI researches company from website + LinkedIn
- Generates likely interview questions
- Voice practice: Cubiqo asks questions, user answers, feedback given

Current vs Ideal Reality

Step	Ideal	Current Code	Gap
Resume upload	PDF/DOCX/TXT	✓ Implemented	—
LinkedIn URL	Field exists	✓ In schema	Not validated
Target roles/skills	Full form	✓ Types defined	Setup UI at /job-hunt/setup
Questionnaire	Common questions	✓ API exists	Q&A content not seeded

Step	Ideal	Current Code	Gap
Job search (browser)	6+ platforms	🟡 Browser wired	Auto-search not scheduled
Job scoring	AI match score	❌ Not implemented	Needs LLM scoring step
User review UI	Ranked feed	🟡 Dashboard exists	Feed not built
Auto-apply	Browser fill/submit	🟡 Playwright ready	Submit flow not fully built
Email alerts	Daily + interview	✅ Report types exist	Email send not wired
Interview prep	Voice Q&A	❌ Not started	Future feature
Application tracking	Full status flow	✅ Types + API	Dashboard shows stats

What to Build to Complete Job Hunt

Week 1:

- ❑ Job search automation: schedule browser agent to search LinkedIn/Indeed using user's target roles + locations at configurable intervals
- ❑ Job scoring: LLM call to rate each job against resume (0-100 match score)
- ❑ Jobs feed UI: ranked list with match score, apply/skip buttons

Week 2:

- ❑ Auto-apply flow: complete the form-fill + submit sequence
- ❑ Wire daily email report via Resend
- ❑ Interview alert: detect "interview" in email subject → send notification

Week 3:

- ❑ Company research agent: before application, A7 researches company → cultural fit analysis → personalised cover letter
- ❑ Interview prep: AI asks role-specific questions → user answers via voice → AI gives feedback on content + communication

Topic 9 — Daily Journal: Full User Flow & What's Pending

Current Working Flow (Confirmed in Code)

Step 1: User goes to /journal

- App checks: has user journaled in the last 24 hours?
 - ✅ API: GET /api/journal?sessionId={id}&userId={uid}
 - ✅ Response: { canJournal: true/false, nextAvailable: ISO timestamp }
- If already journaled: shows "Come back tomorrow" gate with countdown
- If can journal: shows "Begin today's session"

Step 2: Journaling flow (/components/journal/JournalFlow.tsx)

- ✅ 8 guided prompts, BigBoss confessional style:
 1. "How are you feeling right now? Really feeling?"
 2. "What happened today that actually mattered?"
 3. "How did that make you feel?"
 4. "What color is today – Green/Yellow/Red/Orange?"
 5. "What did you learn?"
 6. "What's one thing you want to accomplish tomorrow?"
 7. "Big picture – where are you heading in the next month?"
 8. "Anything else you need to get off your chest?"
- ✅ User types response to each prompt

- ✓ "Next" advances through prompts
- ✓ Final prompt saves to database

Step 3: Save & Summary

- ✓ POST /api/journal/entries – creates entry with:
 - session_id, user_id
 - all prompt responses
 - color_category (Green/Yellow/Red from prompt 4)
 - duration_seconds
- ✓ Email summary queued for user after completion
- ✓ Analytics event tracking

Step 4: History

- ✓ GET /api/journal/history – paginated, searchable
- /journal/history page exists but is NOT connected to API
- ✓ GET /api/journal/stats – streak, total entries, color distribution
- ✓ GET /api/journal/summary – AI-generated weekly summary

What Is Pending for Full Journal Experience

Week 1 – Connect the UI:

- /journal/history page: wire to /api/journal/history
- Show streak counter: "🔥 7-day streak" in journal header
- Calendar heatmap: GitHub-style activity grid showing journal days
- Color distribution chart: what % of days were Green/Yellow/Red

Week 2 – Voice Journal:

- Voice mode in journal: user speaks responses → Whisper transcribes
- Cubiqo reads each prompt aloud via TTS before user responds
- Creates a 15-minute "audio therapy session" feel

Week 3 – Insights & AI:

- Weekly summary email: "This week you felt Yellow 4 times, Red 2 times..."
- Pattern detection: "You feel Red every Monday – notice that?"
- Connection to RGY: journal color auto-updates capsule (patent opportunity #3)
- Memory extraction: key facts from journal → added to Journey Memory

Week 4 – Social / Connection:

- RGY connection: if 3+ consecutive Red days → proactively suggest RGY match with someone in Green (who might be in a position to help)
- Crisis detection: if journal responses contain crisis patterns → immediately surface crisis resources + duo-mode check-in

Topic 10 — RGY Keywords & 3 Intent Types: DB Status

What Exists in Database (Confirmed)

```
-- rgy_capsules table (CONFIRMED in migration 20260218000200):
color    TEXT: 'green' | 'yellow' | 'red' ← COLOR CONFIRMED
intent   TEXT: 'collaborate' | 'trade' | 'company' ← 3 INTENTS CONFIRMED
keywords JSONB[] ← max 50 keywords, GIN indexed ← KEYWORDS CONFIRMED

-- DB Constraint (semantically meaningful):
yellow → intent MUST be NULL (open/neutral state)
```

```

green → intent MUST be 'collaborate' | 'trade' | 'company'
red   → intent MUST be 'collaborate' | 'trade' | 'company'

-- RGY Chat Rooms table (CONFIRMED):
name, color, intent, keywords, geofence, expiry, max_participants

-- API routes (CONFIRMED working):
POST /api/rgy/intents      ← register intent
GET  /api/rgy/opportunities/discover ← find matches
POST /api/rgy/opportunities/express-interest ← signal interest
GET  /api/rgy/subscription ← subscription status

```

The 3 Intents Explained

Intent	Meaning	Example Use Case	Matching Logic
collaborate	Creative/technical partnership	"I want to build something with someone"	Match two GREEN/collaborate users who share keywords
trade	Exchange of services/goods	"I have X skill, need Y skill"	Match complementary keyword sets
company	Build a company together	"Looking for a co-founder"	Match users with complementary domains (tech + biz)

What Is Fully Functional vs Needs Work

FULLY FUNCTIONAL:

- ✓ capsule creation with color + intent + keywords
- ✓ 4-stage matching algorithm (color → intent → keyword → vector)
- ✓ RGYRooms component: browse rooms by color + intent
- ✓ Join room: user can enter a chat room
- ✓ Geofence filtering (Haversine SQL function)
- ✓ Vector cosine similarity (pgvector overlay)

NEEDS WORK:

- Keyword autocomplete UI: suggest keywords as user types
- Capsule editor UI: where does user create/edit their capsule?
(RGYColorSelector.tsx and KeywordPanel.tsx exist – need to be wired into a flow)
- Proactive Cubiqo matching: Cubiqo goes and finds matches without user searching
(see Topic 11)
- RealTime notifications: when a match is found → notify user
- Match quality feedback: "Was this a good match?" for ML improvement
- Journal → capsule color sync (the patent-critical closed loop)

Topic 11 — RGY Chat, Capsule Logic & Proactive Matching

Is the Full System Understood in the Codebase?

YES — the full system is designed and partially implemented.

```

CAPSULE = color × intent × keywords
(e.g., "GREEN × collaborate × [react, typescript, fintech]")

```

WHAT THE UI CONFIRMS:

- ✓ RGYColorSelector.tsx – color selection UI exists
- ✓ KeywordPanel.tsx – keyword entry UI exists
- ✓ RGYIntentKeywordList.tsx – intent + keyword display component
- ✓ RGYRooms.tsx – browse + join chat rooms
- ✓ OpportunityFeed.tsx – opportunity discovery feed component

WHAT THE DB CONFIRMS:

- ✓ rgy_capsules table with all fields
- ✓ rgy_chat_rooms table
- ✓ Staged matching functions in SQL
- ✓ user_intents + opportunities tables (with vector similarity)
- ✓ RGY chat from Emergent platform was pushed (/api/rgy/* routes exist)

Proactive Matching — What Cubiqo Should Do Without User Asking

PROACTIVE FLOW (to be built):

Step 1: Cubiqo reads the user's active capsule

→ color = GREEN, intent = collaborate, keywords = [AI, voice, nextjs]

Step 2: Cubiqo runs the matching algorithm against all active capsules

→ finds 3 potential matches (anonymised at this stage)

Step 3: Cubiqo presents options to user in chat:

"I found 3 people who might align with what you're building.
One is a designer (GREEN × collaborate × [figma, ui, saas]).
Want me to find out if they're interested?"

Step 4: If user says yes → Cubiqo sends anonymous signal to matched user

→ POST /api/rgy/opportunities/express-interest (with anonymous=true)
→ Matched user gets notification: "Someone with aligned interests found you"

Step 5: Both users consent to reveal → Chat room created

→ Room expires in 7 days if no activity
→ If both parties join → connection formed → may or may not reveal identity

WHAT IS ALREADY BUILT FOR THIS:

- ✓ /api/rgy/opportunities/discover endpoint
- ✓ /api/rgy/opportunities/express-interest endpoint
- ✓ Anonymisation: can be done at API layer before revealing user IDs
- ✓ Chat rooms: auto-created with expiry

WHAT NEEDS TO BE BUILT:

- Cron: /api/cron/rgy-discovery already exists! Schedule on Vercel
- Notification: when a match is found → push notification to user
- Consent flow: before identity reveal, get explicit consent from both
- Chat UI: integrate RGY room chat into main Cubiqo chat interface
- UI connection: make the RGY section visible from /chat sidebar

How to Kickstart RGY (Immediate Steps)

Week 1 – Make capsule creation easy:

- Add "Set your RGY Status" to onboarding flow (after first voice message)
- Simple 3-step capsule creation: pick color → pick intent → add keywords
- Show active capsule in chat header as a colored dot

Week 2 – Enable proactive matching:

- Activate `/api/cron/rgy-discovery` on Vercel Pro
- Add notification when match found: in-app banner + email
- Wire the consent flow before identity reveal

Week 3 – Show the feed:

- Add "RGY tab" in main navigation
- Show: your capsule | matches found today | rooms you're in
- Chat rooms: integrate into existing voice/chat interface

Topic 12 — Where to Get Test Users + First 1,000 Users

Getting Test Users (Before Launch)

PHASE 0 – Friends & Founders (0-50 users):

- Slack: personal network, founder communities
- WhatsApp: close circle + extended network
- Give them direct access URL + Founders Pass
- Conduct 10 structured user interviews (30 min each via Zoom)
Ask: What did they try first? Where did they get stuck? What surprised them?

PHASE 1 – Community Seeding (50-200 users):

- Reddit:
r/Entrepreneur, r/solopreneur, r/productivity, r/artificialintelligence
Post: "I built a voice-first AI OS that runs on YOUR API keys – here's a demo"
Not promotional – lead with a genuine experiment or finding
- Hacker News: Show HN post – leads with technical credibility
"Show HN: CubiQo – a voice AI companion with BYO API key encryption"
- Product Hunt upcoming page: collect upvote pledges before launch
- IndieHackers: post as a build-in-public story
- Typefully/Tweethunter: schedule 30-day Twitter/X thread series
Day 1: "I'm building a WeChat for AI..."
Day 5: "Here's why client-side AES-256 encryption matters for AI privacy"
Day 10: "Our voice AI detects your emotional tone and adjusts its voice parameters"
(Build-in-public creates organic followers who convert to users)

PHASE 2 – Creator Partnerships (200-500 users):

- Target YouTube channels: Ali Abdaal, Thomas Frank, Tiago Forte, Mike Peralta
→ Offer 6 months free Pro in exchange for honest video review
→ Audience: productivity-obsessed professionals (exact ICP)
- Target AI newsletters: TLDR AI, The Rundown, Superhuman
→ Offer exclusive beta access to their reader list
- Target LinkedIn creators: Lara Acosta, Justin Welsh (solopreneur audience)
→ Use Social Army to prepare personalised outreach for each

PHASE 3 – Paid Acquisition (500-1000 users):

- Start ONLY after Day-30 retention > 20% (otherwise you're wasting money)
- Channels: X/Twitter Ads (tech/AI audience), LinkedIn Ads (professional persona)
- Ad creative: 15-second screen recording of 3D cube morphing as user speaks
→ "This is your new AI OS. It runs on your API keys."
- Landing page A/B test: cube demo vs. problem statement vs. social proof

BEST PLATFORMS TO RECRUIT TESTERS:

- Betalist.com – pre-launch testers who specifically sign up for betas
- BetaFamily.com – engaged testers who give structured feedback
- TestFlight (iOS) / Google Play beta – for mobile future

- Slack communities: Online Geniuses, Product Hunt Makers, Indie Founders
- Discord servers: Midjourney, Notion, Obsidian communities

Topic 13 — Milestones Aligned to Social, Launch & Growth

The Full Timeline

Milestone	Target Date	Social Signal	Technical Gate
Patent provisionals filed	Day 7 from now	—	File before any public post
P0 sprint complete	Day 14	Soft tweet: "Billing is live"	Stripe + ToS + caps fixed
Waitlist open	Day 14	Twitter/LinkedIn: "We're accepting 100 beta testers"	/waitlist page
50 beta users	Week 3	Reddit + HN Show HN post	Feature flags set for beta
10 user interviews done	Week 4	Build-in-public post: learnings thread	Record + publish findings
Day-30 retention measured	Week 6	—	Analytics ≥10 events live
\$1K MRR	Month 2	Tweet: "First \$1K MRR 🎉"	Stripe verified
Product Hunt launch	Month 3	Full PH campaign	After provisionals + billing
Hacker News Show HN	Month 3	Technical deep-dive article	After PH launch
\$5K MRR	Month 4	LinkedIn post: "from 0 to \$5K MRR in 120 days"	CubiKey portal live
Angel conversations	Month 4-5	—	\$5K MRR + metrics deck
Social Army activated	Month 4	First content wave live	Review gate confirmed
First affiliate partner	Month 4	—	Referral programme live
Accelerator applications	Month 5	—	Metrics + traction proof
\$20K MRR	Month 6	"Crossing \$20K MRR 🚀"	Enterprise accounts
Seed round close	Month 9	—	Post-accelerator

Topic 14 — Who Are Cubiqo's Users

Primary User Persona: The Solopreneur Operator

PERSONA 1 – "The Solopreneur Operator" (highest value, highest fit)

Age: 27-42
Works: Freelancer, consultant, indie maker, content creator
Income: \$4,000-\$20,000/month (unpredictable)
Tools today: Notion, ChatGPT, Gmail, Calendly, LinkedIn, Stripe
Pain: "I run 10 apps and none of them talk to each other.
I forget context between sessions. I can't automate without a team."
What they'll pay for: \$29/month (2 hours of saved work at \$50/hr)
Aha moment: "It remembered what I told it 3 weeks ago"
Entry feature: Voice + Journal
Upsell path: Pro → Agents → Social Army

Secondary Personas

PERSONA 2 – "The Frustrated Developer"
Age: 24-35 | Works: Startup developer, side-project builder
Pain: "ChatGPT is too locked down. Claude is expensive. I want one interface."
Entry: CubiKey BYO mode
Aha moment: "I'm running 7 models through one UI for \$0.10"
Upsell: CubiKey API subscription for their projects

PERSONA 3 – "The Social Media Hustler"
Age: 20-32 | Works: Content creator, personal brand, online coach
Pain: "Posting across 10 platforms is exhausting. Hiring a VA is expensive."
Entry: Social Army (Commander tier)
Aha moment: "It posted 50 pieces of content this week without me touching it"
Upsell: Commander tier upgrade

PERSONA 4 – "The Job Seeker in Transition"
Age: 25-45 | Works: Career pivoter, tech professional, laid-off worker
Pain: "Job hunting is a full-time job. I'm applying to 50 places manually."
Entry: Job Hunt feature (free tier allows limited applications)
Aha moment: "It applied to 20 jobs last night while I slept"
Upsell: Pro tier for unlimited applications + priority alerts

PERSONA 5 – "The Emotionally Exhausted Professional"
Age: 28-50 | Works: High-pressure professional, remote worker
Pain: "I want to process my day but can't afford a therapist."
Entry: Journal (free, always accessible)
Aha moment: "It noticed I felt red 4 Mondays in a row and asked if I'm okay"
Upsell: Pro for voice journaling + companion mode

Topic 15 — MO's Insider Tricks to Make This Mainstream

These are not in the codebase. These are strategic insights from pattern recognition across comparable AI products.

The 7 Insider Moves

Trick 1: Lead with the Cube, Not the Features

The 3D plasma cube morphing as the user speaks is the most emotionally compelling thing about Cubiqo. Every marketing piece should open with a 10-second screen recording of the cube reacting to voice. No competitor has anything like it. This is your hook. The features explain later.

Trick 2: "It remembered what you told it 3 weeks ago"

The single most powerful thing that will make Cubiqo feel different from ChatGPT is persistent contextual memory. The Aha moment is when a user mentions something offhand, comes back 3 weeks later, and Cubiqo references it unprompted. Build the memory highlight explicitly — after each session, show the user "Here's what I learned about you today" in 1-3 bullets.

Trick 3: Make BYO Mode the LEAD, Not the Footnote

Every AI product is leaking user data to their servers. BYO mode with client-side AES-256-GCM encryption is a genuine differentiator. The message: "Your API key never leaves your device." Position this as the privacy-first choice. Privacy-conscious users are the most loyal and highest LTV. They will evangelize for you because it aligns with their identity.

Trick 4: The RGY Colour System as a Social Identity

People love colour-coded identity systems (Red/Blue politics, MBTI, Hogwarts houses). RGY gives users a new social identity: "I'm a GREEN collaborator in the AI space." Build shareable RGY capsule cards — a simple image that says "🟢 Collaborate x AI x Voice x NextJS" that users can post to LinkedIn. It creates curiosity about Cubiqo without explaining it.

Trick 5: Ride the "Agentic AI" Wave Before It Peaks

The market narrative in 2026 is shifting from "chatbots" to "agentic AI" — AI that actually does things, not just talks. Cubiqo's browser automation, job hunt, and Social Army are all agentic. The framing: "Cubiqo doesn't just answer questions. It sends emails, applies to jobs, and posts content while you sleep." This narrative is 6 months ahead of mainstream adoption. Get there before the crowded market does.

Trick 6: The \$0 CAC Flywheel

Every user who uses Social Army to post on LinkedIn/Twitter is also indirectly marketing Cubiqo. Build the Social Army content templates to include: "Made with Cubiqo" watermark (optional but incentivized with extra free posts). Estimate: if 100 Commander users each post 5 times/week across 3 platforms, that's 1,500 brand impressions/week from users, not your marketing budget. This is the WeChat flywheel — the product markets itself through use.

Trick 7: The "Voice Diary" Emotion Hack

Research shows that the most retained app categories are health trackers, note-taking apps, and social apps — because they become part of daily routine. The Journal feature, especially in voice mode, creates a daily ritual. The secret: after the first voice journal, send an email: "Your voice note from today is waiting. Want to hear what Cubiqo noticed?" Play back a 30-second AI summary of what the user said. This creates a loop: journal → playback → insights → journal again. This is more habit-forming than any gamification system.

Summary: What to Build Next (Prioritised by Strategic Impact)

Priority	What	Why	Who	Weeks
🔴 P0	Stripe billing + ToS	Existential — no revenue without this	Blossom	1-2
🔴 P0	File patent provisionals	Public launch destroys rights	Founder	1
🔴 P0	Spending caps → DB	Financial risk in prod	Blossom	1
🟡 P1	Analytics ≥ 10 events	Cannot steer without data	Blossom	2

Priority	What	Why	Who	Weeks
🟡 P1	Journal history UI connected	Primary retention feature	Bubbles	1
🟡 P1	Duo Mode backend wired	Key differentiated feature	Blossom	2
🟡 P1	RGY capsule creation in onboarding	Starts network effect	Bubbles	1
🟡 P1	sitemap.xml updated + llms.txt	SEO / AI SEO	MO	0.5
🟡 P2	Social Army review gate	Legal/reputational safety	Blossom	1
🟡 P2	Job hunt search automation	Key upsell trigger	Blossom	3
🟡 P2	RGY proactive matching cron	Network effect engine	Blossom	2
🟡 P2	Emergent platform MVP	Launch tool for other products	Blossom	4
🟢 Post	CubiKey portal	Revenue Engine 2	Blossom + Bubbles	8
🟢 Post	Referral programme	\$0 CAC flywheel	Blossom	3
🟢 Post	Voice journal mode	Retention + daily ritual	Blossom + Bubbles	3
🟢 Post	Interview prep feature	Job Hunt completeness	Blossom	3
🟢 Post	Blog (3 seed articles)	SEO + AI SEO	MO + A3	2
🟢 Post	Social Army 10×10×10	Viral distribution	Blossom	4

Appendix E — Extended Strategy: 10 Topics + Flagship Spec Gap Analysis

Topic 1 — Tools for Success + 3rd-Party Agencies

1.1 Core Digital Tools Stack

Category	Tool	Purpose	Monthly Cost	Priority
SEO & Keyword Research	SEMrush	Keyword tracking, backlink audit, competitor research, content gap	~\$120/mo (Pro)	P0
Web Analytics	Google Analytics 4	Page views, user journeys, conversion funnels, audience segments	Free	P0
Session Recording	Microsoft Clarity	Heatmaps, session replays, rage-click detection	Free	P0
Product Analytics	PostHog (self-hosted)	Feature flags, funnel analytics, cohort analysis, A/B testing	Free (OSS)	P0
Event Pipeline	Segment.io	Route analytics events to GA4 + PostHog + email tools from one SDK	Free up to 1K MAU	P1
Email Marketing	Resend + React Email	Transactional + drip sequences (already partially wired in codebase)	~\$20/mo	P0

Category	Tool	Purpose	Monthly Cost	Priority
CRM & Lifecycle	HubSpot Starter	Lead capture from landing page, lifecycle stages, deal pipeline	~\$20/mo	P1
Social Scheduling	Buffer or Publer	Schedule posts from Social Army output across 10 platforms	~\$15/mo	P1
Video Creation	CapCut + ElevenLabs	Short-form video content; AI voiceover matching Cubiqo's voice engine	~\$22/mo	P1
Heatmaps (paid)	Hotjar	More powerful session analytics once user base grows	~\$32/mo	P2
A/B Testing	Vercel Edge Config + PostHog flags	Landing page variant tests (plasma vs silver vs tech wireframe)	Included	P1
Uptime Monitoring	BetterStack (Uptime)	Alert on downtime, track 99.9% SLA	Free tier sufficient	P0
Error Tracking	Sentry	Catch runtime errors before users report them	Free up to 5K events	P0
AI SEO Content	Surfer SEO	Optimize blog posts and landing copy for search intent clusters	~\$89/mo	P2
Link-in-bio	Linktree Pro	Central hub for all social profiles pointing to cubiqo.com/join	~\$9/mo	P1

Minimum viable stack cost (P0 tools only): ~\$140/month.

1.2 3rd-Party Agencies & Partners

A. Affiliate Platform (Enable Revenue-Sharing)

Platform	Why	Revenue Model	Setup Time
Impact.com	Industry gold standard; 75K+ brands; supports SaaS payouts	% of MRR attributed to referral	2-4 weeks
ShareASale	Cheaper, simpler; good for digital products	Flat CPA or % commission	1 week
PartnerStack	Purpose-built for SaaS; native Stripe integration	Tiered commission based on MRR	1-2 weeks

Recommendation: Start with **PartnerStack** (SaaS-native, Stripe-native, lowest friction). Graduate to Impact once you have 50+ active affiliates.

What this unlocks: Users who refer others get CubiKey credits or cash. Creators who embed your widget get 20-30% recurring commission. This is the WeChat mini-program parallel.

B. Performance Marketing Agency

You don't need a full agency yet. Use these instead:

Resource	What For	Cost
Mayple.com	AI-matched freelance media buyers; pay per performance	Project-based
Fiverr Pro (verified)	Meta Ads / TikTok Ads setup for a specific campaign	\$200-500 per campaign
GrowthMentor.com	1:1 session with growth marketer who's scaled a similar product	\$50-100/hr

C. Investor Access

Service	What It Does	Cost
Angellist Syndicates	List your raise; angels invest in syndicates of \$1K-10K each	5% carry
Wefunder	Regulation CF crowdfunding; get 500+ micro-investors from your own community	3-5% fee
Gust.com	Submit to accelerators and angel networks en masse	Free
Republic.co	Consumer-facing investment raises; good for brands with a community story	6% fee
DocSend	Share pitch deck + track who read it, how long per slide	~\$45/mo

Solopreneur reality: Your first \$150K-250K will come from your personal network + one or two angels who believe in you, not agencies. Use Wefunder to run a community round once you hit 500 active users — this turns users into stakeholders and creates viral word-of-mouth.

D. Legal & Compliance

Service	What For	Cost
Clerky.com	Delaware C-Corp formation, cap table, SAFEs	~\$800 one-time
Stripe Atlas	Quick incorporation + Stripe account in one flow	\$500 one-time
Termly.io	GDPR/CCPA compliant ToS and Privacy Policy generator	~\$30/mo
WithKawazu.com	AI-assisted software IP attorney (startup rates)	~\$200/hr

Topic 2 — Domain Strategy: cubiqo / coqo / ciqu + 30+ Country TLDs

2.1 What You Have (Assumed Asset Inventory)

Based on the brand:

Domain Set	Examples	Strategic Value
Core Brand	cubiqo.com, cubiqo.ai, cubiqo.io	Primary; highest authority
Phonetic alternates	coqo.com, ciqu.com, cubiqu.com	Brand protection; prevent typosquatting
Country ccTLDs	cubiqo.ca, cubiqo.co.uk, cubiqo.de, cubiqo.in, cubiqo.au, cubiqo.fr, etc.	Geo-targeting, local SEO, legal jurisdiction

Domain Set	Examples	Strategic Value
New gTLDs	cubiqo.app, cubiqo.chat, cubiqo.tech	Product-specific sub-brands

2.2 Recommended Strategy (Priority Order)

TIER 1 – PROTECT (register immediately if not done)

cubiqo.com	← PRIMARY (all traffic lands here)
cubiqo.ai	← AI product credibility signal
cubiqo.io	← Developer/tech audience fallback
coqo.com	← Typo protection
ciqo.com	← Typo protection

TIER 2 – REDIRECT (buy, 301-redirect to cubiqo.com)

cubiqo.ca	→ cubiqo.com	(Canadian market, your home)
cubiqo.co.uk	→ cubiqo.com	(UK – biggest English market)
cubiqo.in	→ cubiqo.com	(India – massive AI market)
cubiqo.au	→ cubiqo.com	(Australia – Anglophone)
cubiqo.de	→ cubiqo.com	(Germany – GDPR-forward)
cubiqo.fr	→ cubiqo.com	(France – EU anchor)
cubiqo.app	→ cubiqo.com	(Mobile installs)

TIER 3 – FUTURE (hold, activate only at scale)

cubiqo.ae	– UAE launch (high AI adoption)
cubiqo.sa	– Saudi Arabia
cubiqo.sg	– Singapore (APAC hub)
cubiqo.br	– Brazil (fastest-growing AI market 2025)
cubiqo.mx	– Mexico (Latin America gateway)

2.3 Technical Implementation

Action	How	Impact
301 redirects	Vercel → Project Settings → Redirects, or Cloudflare Page Rules	Preserves link equity; all ccTLD authority flows to cubiqo.com
Hreflang tags	Add <code><link rel="alternate" hreflang="en-ca" href="..."></code> in <code>src/app/layout.tsx</code>	Signals Google to show correct URL to each country
Geo-targeting in GSC	Set cubiqo.com as international target in Google Search Console	Tells Google your site serves all markets
CDN edge locations	Vercel Edge Network handles this automatically	Serves from nearest edge to each user

2.4 The CQ/Coqo Brand Option

If you want a shorter consumer brand (like how Instagram → Meta), consider:

- **coqo.ai** as the consumer-facing social/RGY app brand
- **cubiqo.com** as the platform/enterprise brand
- This mirrors Google/YouTube, Facebook/Instagram, Salesforce/Slack

Cost to hold 30 domains: ~\$900/year. Worth every dollar for brand protection at this stage.

Topic 3 — Landing Page: Conversion Standards + Silver Wireframe Feedback

3.1 Industry Conversion Benchmarks

Landing Page Type	Median CVR	Top Quartile CVR	What "Convert" Means
SaaS free trial	2-5%	8-12%	Email sign-up
AI tool waitlist	8-20%	25-40%	Join waitlist
Consumer app (curiosity hook)	12-35%	40-60%	Tap to enter + complete onboarding
Product Hunt launch day	2-6%	10-15%	Upvote + sign-up

Your current landing does NOT have a clear CTA above the fold. "Tap to begin" only shows up after 3 seconds of animation. This is the single biggest conversion leak.

3.2 What the Industry Standard Landing Page Has

ABOVE THE FOLD (visible without scrolling, no animation required):

LOGO + wordmark (top left)

HERO HEADLINE – "What it does in 8 words"

SUB-HEADLINE – "Who it's for and what changes for them"

[PRIMARY CTA BUTTON] "Start Free" / "Join Waitlist"

Social proof: "12,000 people already using this"

VISUAL HOOK (cube / animation / screenshot)

BELOW THE FOLD:

Pain → Solution narrative

3 feature cards with icons

Testimonials / logos

FAQ

Second CTA

3.3 Silver Wireframe — MO's Design Feedback

The silver wire aesthetic is **the right direction**. Here is why and what to tweak:

Element	Current (Plasma Wave)	Silver Wire (New)	Verdict
Emotional tone	Mystical, spiritual	Precise, intelligent, premium	✅ Better for B2B + sophisticated users
Color temperature	Warm purple/cyan	Cool silver-chrome	✅ Signals "intelligence"
Animation speed	Fast plasma pulse	Slow elegant morph	✅ More premium feel

Element	Current (Plasma Wave)	Silver Wire (New)	Verdict
Background	Near-black + purple glow	Near-black + blue-silver glow	✓ Cleaner
Text legibility	White on dark — good	White-silver on dark — good	✓
CTA visibility	Hidden below fold	Still hidden — needs fix	✗ Fix this

Recommended tweaks to the silver cube landing:

1. **Add a CTA button overlay** — don't rely on "tap anywhere". Add:
`[ Start Free — No Card Required ]`
as a pill button below the tagline. Keep "tap to begin" as secondary text.
2. **Add a single social proof line** between the wordmark and the tagline:
`Trusted by 1,200+ solopreneurs ← update this as your numbers grow`
3. **Morph timing** — the current 4-second cube→sphere cycle is excellent. Keep it.
4. **Add a subtle scanline/grid overlay** at 3-5% opacity to reinforce the "intelligence dashboard" aesthetic.
5. **The corner decorative marks** (already in the code) are perfect — don't remove them. They signal precision.

A/B Test Plan:

```

Variant A: plasma-wave (current default)
Variant B: silver-wireframe (new)
Variant C: silver-wireframe + CTA button overlay

Metric: email capture rate on the landing → app entry
Tool: PostHog feature flags + Vercel URL params (?landing=silver-wireframe)
Duration: 14 days minimum, 500+ sessions per variant

```

Topic 4 — RGY Score / CQ Intelligence (CQ Score Design)

4.1 What Exists Today (Code Reality)

- From `supabase/migrations/` and `src/lib/`:
- RGY zones are assigned per journal entry / chat intent
 - Keywords are tracked with zone colour + intent type (collaborate/trade/connect)
 - There is no "CQ Score" numeric value — only zone labels

4.2 The CQ Score Vision (Design Proposal)

CQ = Cubiqo Intelligence Quotient — a living, composite score that represents a user's **growth trajectory across all life dimensions**, not a judgment on where they are.

CQ SCORE ARCHITECTURE

```

CQ SCORE (100-999)
|
|— DIMENSION 1: Clarity (RED zone activity)
|   "How much is actively changing in your life?"
|   Score inputs: journal frequency, topics flagged as
|   high-change, emotional variance across entries
|   Range: 0-333
|
|— DIMENSION 2: Growth (YELLOW zone activity)
|   "How actively are you learning & connecting?"
|   Score inputs: new keywords, new intent types explored,
|   RGY matches made, job hunt activity, courses taken
|   Range: 0-333
|
|— DIMENSION 3: Alignment (GREEN zone activity)
|   "How aligned are you with your intentions?"
|   Score inputs: journal consistency, completed goals,
|   RGY connections that led to outcomes, positive sentiment
|   Range: 0-333

BONUS (+1 to +99):
  Connection bonus: friends added, messages exchanged
  App engagement: streak days, features used

```

Key principle (from your brief):

"Red should not be derogatory — it is change, it is dynamism, it is life in motion."

The scoring system should **reward change**, not penalise it:

- A user deeply in RED (lots of change) should have a high CQ if they're journaling about it
- A user stuck in YELLOW forever with no growth should score lower than a dynamic RED user

4.3 ConstantQuestioner.com Integration

Domain tie-in: `constantquestioner.com` → the "CQ number" concept.

Concept	How It Works
CQ Score	Numerical 100-999, visible on profile
CQ Number	Already implemented (CQ-XXXX-XXXX format in <code>src/lib/cq-to-cq/cq-number-generator.ts</code>)
Tie the two	CQ Score is displayed alongside CQ Number on the user profile card
ConstantQuestioner	A content brand / quiz platform that feeds CQ Score inputs: complete a questionnaire, your answers adjust your CQ Score dimensions

Database changes needed:

```

-- Add CQ Score to user profiles
ALTER TABLE profiles ADD COLUMN cq_score INTEGER DEFAULT 100;
ALTER TABLE profiles ADD COLUMN cq_clarity INTEGER DEFAULT 0;
ALTER TABLE profiles ADD COLUMN cq_growth INTEGER DEFAULT 0;
ALTER TABLE profiles ADD COLUMN cq_alignment INTEGER DEFAULT 0;

```

```
ALTER TABLE profiles ADD COLUMN cq_last_calculated TIMESTAMPTZ;

-- CQ Score history for charting
CREATE TABLE cq_score_history (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
  cq_score INTEGER NOT NULL,
  cq_clarity INTEGER,
  cq_growth INTEGER,
  cq_alignment INTEGER,
  calculated_at TIMESTAMPTZ DEFAULT NOW()
);
```

API endpoint needed: `POST /api/cq/recalculate` — triggered after each journal entry or RGY match event.

4.4 Why This Is Powerful

1. **Gamification without toxicity** — the score grows with engagement, never punishes users
2. **Social identity** — "I'm a CQ-847" becomes a badge, like credit scores but positive
3. **Investor metric** — average CQ score growth over time = proof of product value
4. **Monetization hook** — "CubiKey Pro users score 34% higher CQ on average" — upgrade incentive

Topic 5 — CQ Number as Universal Identity: Permanent vs Rotating + Adoption Strategy

5.1 What Already Exists

In `src/lib/cq-to-cq/cq-number-generator.ts`:

```
Format: CQ-XXXX-XXXX (e.g., CQ-8F3A-2K9B)
Rotation: every 30 days (privacy feature)
Entropy:  $32^8 = \sim 1$  trillion combinations
QR code: generateCQNumberQRData() already implemented
Shareable link: https://cubiqa.com/add/CQ-XXXX-XXXX already implemented
```

5.2 The Problem: Rotating $\neq$ Identity

A rotating number is **great for privacy** but **terrible for identity**. Phone numbers don't change. Instagram handles don't change. For CQ Number to work as a persistent identity anchor, you need a two-layer system:

```
LAYER 1 — Permanent CQ Handle (like @username)
  CQ-8F3A-2K9B ← permanent, assigned at account creation, never changes
  Used for: profile sharing, QR codes, "find me" links
  Stored as: immutable in profiles.cq_handle column

LAYER 2 — Rotating Privacy Token (already exists, keep it)
  CQ-2K9B-TEMP ← rotates every 30 days
  Used for: anonymous RGY matching, temporary connections
  Stored as: cq_numbers table (already implemented)
```

Code change needed: Add `cq_handle` column to profiles as immutable; generator runs once at registration.

5.3 Why Would People Want a CQ Number? (Adoption Psychology)

The same reason people want a LinkedIn URL, an Instagram handle, a Linktree. The answer is always: **social proof + access**.

Incentive	Mechanic	Analogy
Exclusivity	First 10,000 users get "Founder CQ" prefix: <code>CQ-F-XXXX</code>	Discord Nitro early badge
Network effect	"Add me on Cubiqo: CQ-8F3A-2K9B" becomes a social phrase	Cash App \$cashtag
Verification	CQ-Verified badge for users with 90-day+ accounts + CQ Score > 500	Twitter/X blue tick
Commerce	"Share your CQ Number, your friend gets 10% off their first CubiKey"	Referral codes
Professional	Add CQ Number to LinkedIn bio, email signature	Personal website URL
Business card	QR code of CQ Number on physical card → your Cubiqo profile	WeChat QR code
Emergency protocol	"In crisis? Share your CQ Number with someone you trust — they can see your zone"	ICE contact

The WeChat parallel: In China, you don't exchange phone numbers, you exchange WeChat QR codes. CQ Number is the same play — but AI-native.

5.4 Implementation Checklist

- Add `cq_handle` (permanent) to profiles table — migration needed
- Generate `cq_handle` at account creation, never change it
- Show CQ Number prominently on profile page (above the fold)
- Add "Copy CQ Number" button with one-tap clipboard copy
- Add QR code display on profile (`qrcode.react` library, already in many OSS stacks)
- Add "Find by CQ Number" search on friends/connection page
- Add CQ Number to the onboarding flow as a celebratory moment:
"You are CQ-8F3A-2K9B — share this with anyone to connect"
- Add CQ Number to email signature template sent after registration
- Create `/add/[cq-number]` public landing page (already referenced in generator)

Topic 6 — Cubiqo as Commerce Layer / WeChat Super-App

6.1 What Already Exists (Confirmed in Code)

`src/lib/deals/deals-service.ts` — A full Groupon-style deals engine already exists:

```
// Already implemented:
detectInterestCategories(text) // NLP keyword → deal category
hasDealIntent(text)           // "discount", "deal", "save" → trigger
fetchDeals({ category, maxResults }) // returns curated deal catalog
getContextualDeals(userMessage) // full pipeline: detect → fetch → return
```

```
// 10 deal categories already catalogued:  
// food, travel, shopping, entertainment, beauty, fitness,  
// electronics, education, services + more
```

The gap: All deals are hardcoded mock data with `url: '#'`. No real affiliate connection yet.

6.2 WeChat Super-App Layer (What to Build)

CUBIQO SUPER-APP LAYERS

LAYER 1 (EXISTS): AI Conversation + Memory

User talks to Cubiqo → gets smart responses → memory stored

LAYER 2 (EXISTS): RGY Journaling + Peer Matching

User journals → RGY zone classified → capsule created → matched

LAYER 3 (EXISTS - MOCK): Deals & Commerce

User mentions "cheap flights" → deals engine returns offers

NEXT: Connect to real affiliate feeds (Commission Junction, CJ Affiliate)

LAYER 4 (PARTIAL): CQ-to-CQ Messaging & Calls

Friends system, direct messages, WebRTC calls all coded

NEXT: Make this the primary social layer

LAYER 5 (NOT BUILT): Mini-Programs / CubiKey Ecosystem

Third parties build modules that live inside Cubiqo

User pays once, gets curated apps from trusted ecosystem

This IS the WeChat mini-program play

LAYER 6 (NOT BUILT): Payments & Wallet

CubiKey credits as in-app currency

Stripe → CubiKey top-up → spend on deals, mini-programs, upgrades

LAYER 7 (NOT BUILT): Local Services

Geo-aware deals + CQ-matched service providers in your city

"You mentioned you need a therapist → here are 3 CQ-verified therapists near you"

6.3 Affiliate Commerce: How to Actually Monetize the Deals Engine

Phase 1 — Week 1-2: Wire real affiliate feeds

Affiliate Network	Best For	Integration
Commission Junction (CJ)	Travel, electronics, retail	REST API + XML feed
Rakuten Advertising	Fashion, beauty, lifestyle	API
Amazon Associates	Electronics, books	Product Advertising API
Groupon Getaways API	Travel, experiences	REST (requires approval)
Honey / Capital One Shopping	Price comparison	Partner programme

Revenue per transaction: 3–15% commission depending on category. At 1,000 active users making 1 purchase/month averaging \$50 → **\$1,500–7,500/month passive income from deals alone.**

Phase 2 — Month 2: Add deal personalization

Connect deals to RGY zone + journal keywords. If user's GREEN zone keywords include "travel" and "wellness" → surface only travel and spa deals. This is the contextual moat that generic deal sites don't have.

Phase 3 — Month 3: CQ-Verified Local Professionals

Allow therapists, coaches, designers to list their services. Cubiqo takes 15% booking fee. User pays via Stripe. Service provider gets a CQ Badge. This is the LinkedIn-meets-Groupon-meets-Airbnb layer.

6.4 Discount Code Engine (What the User Described)

"it has a code for discount AND IT knows where exactly to buy and it knows the user for a good contextual recommendation"

This is **exactly** the deals engine but with two additions:

1. **Promo code field** in the deal object (add `promoCode?: string` to the Deal type in `src/lib/deals/types.ts`)
2. **Contextual matching** — already implemented via `detectInterestCategories()`, just needs real data

```
// Proposed enhanced Deal type
interface Deal {
  // ... existing fields ...
  promoCode?: string // "CUBIQ020" for 20% off
  promoExpiry?: string // ISO date string
  deepLink?: string // Direct link with affiliate tracking parameter
  affiliateId?: string // For attribution tracking
  isPersonalized?: boolean // Flag if this was surfaced by context engine
}
```

Topic 7 — Silver Wire Cube: Implementation + Visual Guide

7.1 What Was Built (This PR)

New file: `src/components/SilverWireLandingCube.tsx`

The component implements:

SILVER WIRE CUBE — TECHNICAL BREAKDOWN

GEOMETRY:

- BoxGeometry(1.6, 1.6, 1.6, 3, 3, 3)
- subdivided for smooth morph
- EdgesGeometry(BoxGeometry(1.6,1.6,1.6)) for crisp lines

MORPH ANIMATION:

- t = 0: Pure cube wireframe
- t = 0.5: Midpoint sphere-ish form
- t = 1: Full sphere (all vertices projected to unit sphere surface)
- Cycle: 4 seconds full cube→sphere→cube via $\sin(t * 0.5)$

Implementation: per-frame Float32Array lerp between box and sphere positions

COLOR PALETTE:

Edges: #D8E4F0 (blue-silver, 0.85 opacity)
Fill mesh: #C8C8D0 (gunmetal silver, wireframe: true, opacity: 0.18)
Corner spheres: #E8F0FF (near-white specular)
Corner emissive: #A8C0FF (very subtle blue glow)
Rings: #B0C8E8 (cool blue-silver, 0.28–0.35 opacity)
Background: #080A0F (near-black, cooler than pure black)

LIGHTING:

Ambient: 0.6 intensity, #D8E8FF tint
Point 1: [4,4,4], intensity 1.8, white
Point 2: [-4,-2,3], intensity 0.8, #A0B8D8
Point 3: [0,-4,-3], intensity 0.5, #C8D8F0

EXTRAS:

3 axis rings (torus geometry) at 0.28–0.35 opacity
8 corner accent spheres (tiny, 0.028 radius)
4 corner L-bracket decorative marks (CSS, not WebGL)
Horizontal rule gradient accent lines (CSS)

7.2 How to Preview It Right Now

Option A — URL parameter (no code change needed):

```
https://your-staging-url.vercel.app/?landing=silver-wireframe
```

Option B — Set as default:

```
# In your .env.local
NEXT_PUBLIC_LANDING_DEFAULT=silver-wireframe
```

Option C — Switch in `src/config/landing.ts` :

```
defaultVariant: 'silver-wireframe' // change from 'plasma-wave'
```

7.3 Design Comparison Table

Property	Plasma Wave (current)	Tech Wireframe	Silver Wire (new)
Dominant hue	Purple / cyan	Blue / pink / orange	Silver / chrome / blue-silver
Energy	High, kinetic	High, electric	Low, meditative, precise
User feeling	"Mystical AI"	"Sci-fi tech"	"Premium intelligence tool"
Brand fit	Early-stage wow	Hacker/developer	Enterprise / discerning user
Animation	Particle storm	Energy pulses	Geometric morph
Load weight	High (120K particles)	Medium (shaders)	Low (simple geometry)
Mobile performance	Medium	Medium	✅ Best (lightweight)
Conversion hypothesis	Wonder → curiosity	Interest → excitement	Trust → action

MO's recommendation: Run the silver wireframe as Variant B in A/B test. If it wins (which I expect it will for your core solopreneur demographic who values intelligence and precision), set it as default.

7.4 Next Enhancements (Post-Launch)

Enhancement	Complexity	Impact
Voice-reactive morph speed	Medium	Cube morphs faster when user speaks
RGY zone coloring	Low	Edges shift to Red/Yellow/Green based on user's current zone
CQ Score pulse	Low	Opacity pulses at rate proportional to user's CQ Score
Mouse parallax	Low	Cube tilts toward mouse cursor for depth illusion
Click ripple	Low	Tap sends shockwave through cube edges

Summary Priority Table

Topic	Key Action	Priority	Owner	Time
Tools	Set up GA4 + Clarity + PostHog + Sentry	P0	Blossom	2 days
Tools	SEMrush subscription + initial keyword audit	P0	JO + MO	1 day
Tools	Resend drip sequence (3-email welcome series)	P0	Blossom	3 days
Agencies	PartnerStack affiliate platform integration	P1	Blossom	1-2 weeks
Domains	301 redirects from all ccTLDs to cubiqo.com	P0	D2	1 day
Domains	Hreflang tags in layout.tsx	P1	Bubbles	0.5 day
Landing	CTA button overlay on silver wire landing	P0	Bubbles	1 day
Landing	A/B test: plasma-wave vs silver-wireframe	P1	Blossom+PostHog	3 days
CQ Score	DB migration for cq_score columns	P1	Guy	1 day
CQ Score	Recalculation API endpoint	P1	Blossom	2 days
CQ Score	Profile page CQ Score display	P1	Bubbles	1 day
CQ Number	Add permanent cq_handle to profiles	P0	Guy	0.5 day
CQ Number	Onboarding celebratory CQ reveal moment	P1	Bubbles	1 day
CQ Number	/add/[cq-number] public landing page	P1	Bubbles	1 day
Commerce	Wire real affiliate API (CJ or Rakuten)	P1	Blossom	1 week
Commerce	Add promoCode field to Deal type	P0	Blossom	0.5 day
Silver Cube	✅ Component built (this PR)	Done	MO	Done
Silver Cube	Add CTA button to silver landing	P0	Bubbles	1 day
Silver Cube	A/B test setup	P1	Blossom	2 days

Section 8 — BYO Mode: The Honest Reality, What True Offline Means, and the Road There

8.1 What BYO Mode Actually Does Today (vs What Was Assumed)

There is a common misconception worth clearing up first.

Dimension	What People Think BYO Does	What It Actually Does
API call destination	Goes directly from user's browser to Anthropic/OpenAI	User's browser → Cubiqo server (/api/chat) → Anthropic/OpenAI (with user's key)
Key storage	Key stays in the user's browser only	Key is AES-encrypted and stored in Supabase profiles.byo_config column
Cloud involvement	None	Cubiqo server is always in the middle — it decrypts and forwards
Cost to user	Billed to their API account	Yes — but still routes through Cubiqo infra
Privacy	Complete	Cubiqo server sees the decrypted key at decrypt-time in memory

Current BYO is a billing model, not a privacy model. The user pays their own cloud bill. Cubiqo still processes the request server-side. This is fine for Phase 1 and is honest — it's stated correctly in the UI ("encrypted before being sent to our backend"). But it is NOT the same as "no cloud call at all."

8.2 Three Tiers of "Bring Your Own" — From Now to True Offline

TIER 1 (DONE – live now)

Browser → Cubiqo Server → Anthropic/OpenAI
(user's encrypted key decrypted here)

✔ User pays own API bill

✗ Cubiqo server still in the middle

✗ Not private from Cubiqo infra

✗ NOT offline

TIER 2 (8–12 weeks – browser-side inference)

Browser → WebLLM (WASM/WebGPU, runs IN browser tab)
Model: Phi-3-mini-4k (2.3 GB download once)
or Gemma-2B-IT (1.8 GB)

✔ Zero server roundtrip for inference

✔ Truly private – model runs on user's GPU

✔ Works offline (after first load)

⚠ Requires WebGPU (Chrome 113+, Edge 113+, no iOS Safari)

⚠ Slower than cloud (3–8 tok/s vs 50+ tok/s)

✗ Actions (email, calendar) still need internet

TIER 3 (16–24 weeks – desktop + mobile native apps)

Tauri Desktop App → Bundled Ollama (llama3.2:3b)

One-click install, no terminal needed

- ✓ Full offline inference, no WASM limitations
- ✓ File system access for local journal, photos, data
- ✓ Can talk to local services (printer, local server)
- ✓ macOS / Windows / Linux
- ⚠ First download ~4 GB (app + model)
- React Native + llama.cpp (iOS/Android)
- ✓ Background inference, native camera, contacts
- ⚠ App Store approval process (2–4 weeks)

8.3 What Already Exists in the Codebase (Useful Foundations)

Asset	File	Status	Notes
Ollama integration	<code>src/lib/ai/ollama.ts</code>	✓ Working	Server-side only; points to <code>localhost:11434</code>
AI router (Ollama-first)	<code>src/lib/ai/router.ts</code>	✓ Working	Routes 80% to Ollama, 20% to cloud
PWA manifest	<code>public/manifest.json</code>	✓ Done	Icons 192/512, display: standalone
Service worker	<code>public/sw.js</code>	✓ Working	Network-first + <code>offline.html</code> fallback
ServiceWorkerRegistration	<code>src/components/ServiceWorkerRegistration.tsx</code>	✓ Working	Registers in production
PWA install prompt	<code>src/components/PWAInstallPrompt.tsx</code>	✓ Working	Android + iOS manual instructions
Offline fallback page	<code>public/offline.html</code>	✓ Done	Basic "you're offline" page
BYO key storage	<code>src/lib/byo/byo-manager.ts</code>	✓ Working	AES-encrypted in Supabase
BYO UI	<code>src/components/byo/BYOSettings.tsx</code>	✓ Working	WCAG 2.1 AA compliant

Key gap: The Ollama route currently only works when Cubiqo's server has Ollama running locally (`localhost:11434`). That is the server's localhost, not the user's machine. To make it truly client-side offline, the model must run in the browser (WebLLM) or in a native app (Tauri + Ollama).

8.4 Tier 2 Implementation Plan — Browser-Side Inference (WebLLM)

Technology: [MLC-AI WebLLM](#) — runs quantised LLMs in browser via WebGPU. No Python, no Ollama, no server.

Step-by-step build plan:

```

Week 1-2: Research + Proof of Concept
- Install: npm install @mlc-ai/web-llm
- Create src/lib/ai/webllm-client.ts
- Wrap WebLLM engine in a React context (useWebLLM hook)
- Gate behind feature flag: feature_flag = 'browser_inference'
- Test with Phi-3-mini-4k-instruct-q4f16_1 (smallest viable model)

```

Week 3-4: UX for model download

- First-launch download modal: "Download 2.3 GB AI model once, run forever offline"
- Progress bar using WebLLM's `initProgressCallback`
- Cache model in IndexedDB (WebLLM handles this automatically)
- Show estimated time remaining

Week 5-6: Feature integration

- Chat route: if user has `browser_inference` enabled AND model loaded → use WebLLM
- Offline detection: `navigator.onLine` → switch to WebLLM automatically
- Memory extraction: run locally (smaller prompt, acceptable quality)
- RGY classification: run locally (keyword-based, already works without cloud)

Week 7-8: Actions bridge

- Email/calendar/social still need internet → show "action queued" when offline
- Sync queue: store pending actions in IndexedDB → execute when back online
- Show user clearly: "Thinking offline ✓ | Will send email when connected ↑"

Browser compatibility gate (must show before offering):

```
const supportsWebGPU = async (): Promise<boolean> => {  
  if (!navigator.gpu) return false;  
  const adapter = await navigator.gpu.requestAdapter();  
  return !!adapter;  
};
```

Browser	WebGPU Support	Notes
Chrome 113+	✅ Full	Best experience
Edge 113+	✅ Full	Same Chromium engine
Firefox	⚠️ Partial	Behind flag, unstable
Safari 17.4+	⚠️ Partial	Metal backend, limited
iOS Safari	❌ None	No WebGPU as of Feb 2026
Android Chrome	✅ Most devices	Depends on GPU

Models to offer (in order of size vs quality):

Model	Size	Speed	Quality	Use Case
Phi-3-mini-4k (q4f16)	2.3 GB	6–10 tok/s	Good	Default choice
Gemma-2B-IT (q4f16)	1.8 GB	8–12 tok/s	OK	Smaller devices
Llama-3.2-3B (q4f16)	2.0 GB	7–11 tok/s	Good	Open weights
Phi-3.5-mini (q4f16)	2.5 GB	5–8 tok/s	Best	Power users

8.5 Tier 3 — Desktop App with Bundled Ollama (Tauri)

Why Tauri over Electron: Tauri uses the OS WebView (WebKit/EdgeWebView2) instead of bundling Chromium. App size is ~10 MB vs ~150 MB for Electron. Rust backend is faster and more memory-efficient.

Architecture:

```
Tauri Rust Core
- Spawn Ollama as a sidecar process
- File system access: journals, photos, local data
- System tray icon: "CubiQo running"
- Auto-updater (built-in Tauri updater)
- Biometric auth (TouchID / Windows Hello)

Next.js frontend (compiled to static)
- All existing React UI unchanged
- AI calls → localhost:11434 (Ollama sidecar)
- Storage → user's local files + SQLite
```

One-click installer experience:

- 1. User downloads CubiQo-1.0.0-setup.exe (Windows) or CubiQo-1.0.0.dmg (macOS)
- 2. Installer places Ollama binary as a Tauri sidecar
- 3. First launch: "Downloading your private AI brain... [██████████] 3.2 GB" (model download)
- 4. Model cached at ~/.cubiqo/models/ — never re-downloads
- 5. All AI inference: local. Internet only for actions (email, web search, social posts)

Timeline: 16 weeks from decision to App Store submission (macOS/Windows). iOS/Android React Native variant: separate 20-week track.

8.6 Immediate Next Steps (What to Do Now)

Priority	Action	Owner	Effort
P0	Add honest copy to BYO UI: "Your inference still routes through Cubiqo servers"	Bubbles	0.5 day
P0	Add "True Offline (Beta)" toggle behind feature flag (disabled by default)	Blossom	1 day
P1	WebLLM PoC: npm install @mlc-ai/web-llm , single chat endpoint test	Blossom	3 days
P1	Enhance service worker: cache chat-related static assets for offline splash	Blossom	1 day
P1	Offline action queue in IndexedDB for emails/posts when offline	Blossom	1 week
P2	Tauri desktop app PoC (Next.js static export + Tauri shell)	MO	2 weeks
P2	React Native shell wrapping current web views	D3	2 weeks

Section 9 — Domain Strategy (cubiqo.ai vs cubiqo.com) + Offline Browser Vision

9.1 cubiqo.ai vs cubiqo.com — The Definitive Recommendation

Short answer: Own both. Primary on .com . Brand on .ai .

Here is the full analysis:

Arguments for .com as primary:

- .com has ~48% of all registered domains — the default expectation globally

- Users type `.com` by muscle memory; `.ai` requires deliberate thought
- Trust signals: enterprise buyers, investors, and press default to `.com`
- Better email deliverability reputation (`@cubiqo.com` > `@cubiqo.ai`)
- SEO: Google treats `.com` with no special advantage, but legacy domain authority tends to concentrate there
- Insurance: if `.ai` ever becomes restricted (Anguilla controls it), `.com` is safe

Arguments for `.ai` as primary:

- Signal: `.ai` immediately communicates what the product is
- Memorability in the AI space: Perplexity.ai, Character.ai, Poe.ai all use it
- Differentiation from generic `.com` SaaS noise
- Premium positioning: `.ai` domains sell for 3-10x `.com` at resale
- Easier to get: cubiqo.com may be harder to secure if already taken

Recommended configuration:

```
Canonical domain:  cubiqo.com           (all SEO juice lives here)
AI brand domain:   cubiqo.ai            (redirects 301 to cubiqo.com)
Alt spellings:     coqo.com / ciqo.com  (redirect to cubiqo.com)
Email:             @cubiqo.com          (primary) + @cubiqo.ai (alias)
```

One exception: If cubiqo.com is NOT owned and cubiqo.ai IS — use `.ai` as primary immediately and try to acquire `.com` over time.

The 30+ country TLD strategy (from previous analysis):

Tier	Domains	Action	Reason
Protect	<code>.com .ai .io .co</code>	Own & keep	Core brand assets
Redirect	<code>.ca .uk .au .de .fr .in .sg .ae</code>	301 → <code>.com</code>	Key markets, affordable
Hold	<code>.sa .br .mx .za .ng .pk .ph .id</code>	Register if <\$50/yr	Emerging markets, option value
Release	All others	Let expire	No ROI, maintenance burden

9.2 The Offline Browser Vision — What Is Actually Feasible

The CEO described: an offline browser that uses AI search, works with user data offline, does photo/video editing, face tagging, AI model comparison, and data synthesis.

Let me split this into three categories: **Can do now**, **Can do in 6 months**, **Fundamentally hard or impossible**.

✔ CAN DO NOW (or within 8 weeks)

1. Offline-cached web pages (reading mode)

- Service worker already exists in `public/sw.js`
- Enhancement: add "Save for offline" button → caches full page HTML in IndexedDB

- User can read saved pages without internet
- Effort: 1 week

2. Local AI search over saved content

- Run keyword + semantic search over IndexedDB-cached pages
- Use existing TF-IDF or embed with a tiny local model (transformers.js, `all-MiniLM-L6-v2`, 80 MB)
- Effort: 2 weeks

3. Photo viewing + basic editing offline

- PWA can access files via File System Access API (Chrome 86+)
- Basic editing (crop, rotate, filter): use `fabric.js` or `jimp` (WASM port)
- No uploads needed — pure client-side
- Effort: 3 weeks

4. AI model answer comparison

- Already partially exists: `router.ts` routes between Claude, OpenAI, Ollama, MiniMax
- UI enhancement: "Compare" button sends same prompt to 2+ providers → side-by-side diff
- Offline: can compare WebLLM models (Phi-3 vs Gemma) without internet
- Effort: 1 week for UI, existing infrastructure handles routing

5. Local journal + memory synthesis

- Journal data is in Supabase BUT can be cached in `localStorage`/IndexedDB on write
- Offline journal: write to IndexedDB → sync to Supabase when back online
- Local synthesis: "Summarise my last 30 days" → WebLLM processes cached journal entries
- Effort: 2 weeks

⚠️ FEASIBLE IN 6 MONTHS (with significant effort)

6. Face tagging in photos

- Technology: `face-api.js` (TensorFlow.js) — runs in browser, 97% accuracy on aligned faces
- Models: `ssd_mobilenetv1` (face detection) + `face_recognition_model` (128D embeddings)
- Workflow: scan photos → cluster by face vector similarity → user names clusters → persistent in IndexedDB
- Limitation: iOS Safari has poor WASM performance; desktop Chrome is excellent
- Effort: 4–6 weeks

7. Video editing (basic)

- `ffmpeg.wasm` runs in browser — can cut, trim, merge, add text overlays, change resolution
- File size limitation: WASM heap is ~2 GB max, so videos >500 MB need chunking
- Real-time preview is slow (not After Effects); good enough for social clips
- Effort: 3–4 weeks for basic editor UI

8. Proactive contextual recommendations (offline)

- User's data (journal, RGY capsules, saved pages) processed locally by WebLLM
 - "Based on your journal this week, you seem to be focused on job hunting — want me to find openings?"
 - Runs as a background Web Worker after WebLLM model is loaded
 - Effort: 4 weeks
-

✗ FUNDAMENTALLY HARD OR NOT ADVISABLE

9. Offline web search (actual search engine)

- A real search index (like Google's) requires petabytes of data — impossible offline
- What IS possible: cached pages + local index of saved content (not "the web")
- Honest positioning: "Search your saved content offline" not "search the web offline"

10. Full video editing (After Effects level)

- GPU-accelerated video processing requires native code (Metal/CUDA) — not available in browser
- WASM is single-threaded-ish; real-time 4K editing is not feasible
- Tauri desktop app with FFmpeg native binary could handle this — separate effort

11. "Sending emails offline"

- SMTP requires internet — no workaround; emails queue locally and send when online
- This is fine and the right UX; just be honest about it in the UI

9.3 The "Cubiqo as AI Browser" Product Vision

Putting it together, here is what a 12-month roadmap for "Cubiqo as AI browser/OS" looks like:

PHASE 1 (Now → Launch): Core AI assistant + memory + PWA

- ✓ Chat, journal, agents, BYO keys
- ✓ PWA install on mobile/desktop
- ✓ Basic offline (cached pages, offline.html)

PHASE 2 (Month 1–3): Offline inference + saved content search

- WebLLM integration (Phi-3-mini in browser)
- "Save page" button → local AI search
- Offline journal with sync queue
- AI model comparison UI

PHASE 3 (Month 4–6): Media tools + face tagging

- Photo viewer + basic editing (crop/filter/text)
- Face clustering via face-api.js
- Video trimmer via ffmpeg.wasm
- Local media library (not cloud upload)

PHASE 4 (Month 7–12): Desktop + mobile native

- Tauri desktop app (macOS/Windows) + bundled Ollama
- React Native mobile app
- Full offline action queue (email/post queued, sent when online)
- Local file system access (journal exports, data backups)

Why this is the right direction — the WeChat analogy revisited:

WeChat succeeded not because it was a great chat app but because it became the infrastructure people lived in. The offline + local-first angle is Cubiqo's version of that:

- Cloud AI (Copilot, ChatGPT, Gemini): always online, always sending your data to servers
 - Cubiqo offline mode: your AI runs on YOUR device, your data stays with YOU
 - That is a genuine competitive moat — especially in privacy-conscious markets (EU, Canada, enterprise)
-

9.4 Implementation Priority Table

Feature	Feasibility	Effort	Revenue Impact	Privacy Story Value
Honest BYO UI copy	✔ Now	0.5 day	None	High (trust)
Offline action queue	✔ Now	1 week	Medium	High
Local journal cache	✔ Now	1 week	Medium	High
WebLLM integration	✔ 8 weeks	6 weeks	Very High	Very High
AI model comparison UI	✔ 2 weeks	1 week	Medium	Medium
Photo editor (basic)	✔ 6 weeks	4 weeks	High	High
Face tagging	⚠ 12 weeks	6 weeks	High	Very High
Video trimmer	⚠ 12 weeks	4 weeks	Medium	Medium
Tauri desktop app	⚠ 20 weeks	16 weeks	Very High	Very High
React Native app	⚠ 24 weeks	20 weeks	Very High	Very High
"Search the web" offline	✖ Not viable	N/A	N/A	N/A

Section 10 — Who Are Our Users?

Grounded in actual Cubiqo feature set: Journal (Rozana), Job Hunt automation, RGY matching, Social Army, Emergent platform, in-app agents, BYO mode, Deals engine, Duo/Companion mode, CQ identity system, voice AI, crisis detection.

10.1 The Core Question — Why It Matters

"Who is the user?" is the most important question in product development. It determines:

- Which features to build next
- Which marketing channels to invest in
- What pricing model works
- What investor narrative to tell
- What the onboarding flow teaches first

Cubiqo is a **multi-persona product** — intentionally so, because it was built as an "AI OS" that different people use for entirely different reasons. That is a strength (broad TAM) and a risk (unfocused messaging). The answer is **to lead with one persona publicly, serve all privately**.

10.2 The Seven User Personas

PERSONA 1 — The Solopreneur Operator

"I run a one-person business and I need a full team inside an app."

Attribute	Detail
Age / Stage	28-45, already running a business or side hustle

Attribute	Detail
Location	North America, UK, Australia, UAE, India (English-first markets)
Market size	64M solopreneurs in US alone; 400M+ globally (World Bank 2024)
Income	\$3K-\$25K/month revenue, under-resourced
Tech comfort	Medium-high; uses Notion, Zapier, ChatGPT, Canva
Core pain	Wears every hat — marketing, ops, client work, admin — simultaneously
Cubiqo features used	Social Army (posts across 10 platforms), Agents (A1-A7 autonomous tasks), Deals engine (finds tools at discount), BYO mode (controls AI cost), Emergent (launches side projects), Journal (BigBoss accountability prompts)
Aha moment	Social Army auto-posts to 10 platforms while they sleep; morning shows 47 new impressions, zero effort
Monetization path	Free → Pro (\$29/mo) within 2 weeks once Social Army value clicks; upgrades to Founders Pass for unlimited agents
Acquisition channel	Twitter/X, Reddit r/entrepreneur, Indie Hackers, Product Hunt
Churn risk	Low — deeply embedded across social + ops workflows
Investor story value	★★★★★ — the "Zapier + Buffer + ChatGPT in one" narrative for the creator economy

PERSONA 2 — The Job Seeker in Transition

"I need a new job and the process is humiliating, manual, and exhausting."

Attribute	Detail
Age / Stage	24-40, recently laid off, career-pivoting, or quietly looking while employed
Location	US, Canada, UK, India, Australia
Market size	200M active job seekers globally at any moment (ILO 2024); US unemployment always ≥ 4M
Income	Variable — either unemployed or \$60K-\$120K employed and frustrated
Tech comfort	Medium; uses LinkedIn daily, spreadsheets to track applications
Core pain	Applies to 50+ jobs manually, hears nothing back, loses track, gets rejected for unknown reasons
Cubiqo features used	Job Hunt automation (resume upload → auto-apply → status tracking), LinkedIn integration, cover letter AI, interview coaching via voice AI, Journal for rejection processing
Aha moment	Uploads resume Monday, wakes up Tuesday to 12 applications submitted + 2 interview alerts in inbox
Monetization path	Free tier for 10 apps/month → Pro (\$29/mo) for unlimited auto-apply; crisis persona is emotionally motivated to pay
Acquisition channel	Reddit r/cscareerquestions r/jobs, LinkedIn job seeker groups, YouTube career coaches, TikTok "job hunt tips"
Churn risk	Medium — leaves when they get a job (natural churn); re-acquires on next career move
	★★★★★ — automated job applications is a hot category; ZipRecruiter is a \$2B company

Attribute	Detail
Investor story value	

Feature gap to fix: Full autonomous application submission (currently tracks but doesn't auto-submit). This single gap is the difference between "interesting tool" and "life-changing product" for this persona.

PERSONA 3 — The Emotionally Overwhelmed Professional

"I am doing okay on paper but I am exhausted, isolated, and have no one to actually talk to."

Attribute	Detail
Age / Stage	27-45, high-functioning but emotionally depleted
Location	Urban North America, UK, Australia — high-cost, high-pressure cities
Market size	76% of US workers report burnout (Deloitte 2023); 1 in 5 adults experience mental health challenge annually
Income	\$60K-\$200K — can afford a therapist but won't commit to one
Tech comfort	Medium; smartphone-heavy, app-native
Core pain	No safe space to process stress; therapy stigma or cost barrier; journaling feels juvenile alone
Cubiqo features used	Journal/Rozana (BigBoss "friend who tells you the truth" prompts), Duo mode (side-by-side companion), voice AI (emotionally modulated responses), crisis detection (self-harm escalation to safe resources), Companion mode
Aha moment	Journal prompt "Your boss just took credit for your work. Say the unsayable." — they type something raw and Cubiqo responds with exactly the right tone: validating, then redirecting
Monetization path	Hardest to convert on price alone → convert via habit (daily journal streak, weekly emotional summary email) → upgrade for voice companion + unlimited history
Acquisition channel	Instagram mental health accounts, Reddit r/mentalhealth r/therapy, therapy-adjacent podcasts, TikTok therapist creators
Churn risk	Very low once habit forms (daily journal creates daily return)
Investor story value	★★★ — mental health tech is a \$6B market but requires careful positioning to avoid regulatory risk; better to position as "emotional productivity" than "therapy"

Critical warning: This persona triggers real crisis situations. The crisis detection escalation path (`policy-router.ts`) MUST be battle-tested before this persona is publicly targeted. Mental health claims in marketing require legal review.

PERSONA 4 — The Privacy-Conscious Tech Enthusiast

"I want powerful AI but I refuse to send my data to Big Tech servers."

Attribute	Detail
Age / Stage	25-45, developer or highly technical non-developer
Location	EU (GDPR-native), Canada, Germany, Switzerland, Japan

Attribute	Detail
Market size	Smaller absolute numbers (~5M globally) but extremely vocal and viral
Income	\$80K-\$200K, willing to pay premium for privacy
Tech comfort	High — runs their own servers, uses Signal, has a VPN, knows what RAG means
Core pain	Every AI product sends their data to OpenAI/Google; no true privacy option exists at a consumer price point
Cubiqo features used	BYO mode (own API key, controls data routing), future WebLLM (runs inference entirely in browser), Ollama integration, privacy-first journal
Aha moment	Sees BYO settings page — "This is the first AI app that lets me use MY key, MY model, and tells me exactly what happens to my data"
Monetization path	Premium/Enterprise tier — this persona pays for the platform fee + will pay for Founders Pass as status symbol
Acquisition channel	Hacker News, r/privacy, r/selfhosted, Mastodon, DEF CON community, Lobsters
Churn risk	Very low — will become an evangelist and pull in 3-5 more users through word of mouth
Investor story value	★★★★★ — EU AI Act compliance + privacy-first positioning is increasingly a competitive moat; powerful for EU expansion narrative

Note: Current BYO still routes through Cubiqo server (only key is user's own). True privacy requires WebLLM (Tier 2, 8 weeks). Do NOT market to this persona as "private" until Tier 2 ships — they will call it out publicly and cause reputational damage.

PERSONA 5 — The Creator / Social Media Entrepreneur

"My content is my product and I can't keep up with the pace of posting."

Attribute	Detail
Age / Stage	20-38, full-time or aspiring creator / influencer
Location	US, UK, Brazil, India, Philippines, Nigeria
Market size	50M+ content creators globally (SignalFire 2023); 2M+ earn full-time income from content
Income	\$1K-\$30K/month from content; time-constrained and inconsistent
Tech comfort	High for social tools; medium for dev tools
Core pain	Content creation is a daily grind; one person cannot maintain 5+ platforms without burning out
Cubiqo features used	Social Army (10 platforms × 10 accounts × scheduled posting), GFXToolz integration (images/videos), AI caption generation, persona management (different voice per platform), Deals (sponsored content deal tracking)
Aha moment	Sets up Social Army once, schedules a week of posts across Instagram/TikTok/Twitter/LinkedIn/YouTube — all from one dashboard
Monetization path	Pro immediately (\$29/mo is trivial if it saves 10 hours/week); potential Founders Pass for white-label client management
Acquisition channel	YouTube creator economy content, CreatorIQ, Creator Commerce, Beehiiv newsletters for creators

Attribute	Detail
Churn risk	Medium — churns if Social Army hits API platform restrictions (LinkedIn 1/day, Instagram approval delays)
Investor story value	★★★★★ — creator economy is the hottest B2C market; Buffer is worth \$80M, Hootsuite \$750M; Cubiqo is Buffer + AI agents

PERSONA 6 — The Developer / Technical Builder

"I want to build and ship products faster without managing 20 different tools."

Attribute	Detail
Age / Stage	22-40, freelance dev, indie hacker, startup engineer
Location	Global (English-first); strong in India, Eastern Europe, Southeast Asia, Canada
Market size	28M developers globally (Evans Data 2024); 4M freelance devs in US
Income	\$40K-\$200K; cost-sensitive on tools, but pays for 10x productivity gains
Tech comfort	Very high — daily VS Code, GitHub, terminal users
Core pain	Too many disconnected tools: IDE + hosting + secrets management + deployment + monitoring = cognitive overhead
Cubiqo features used	Emergent platform (Monaco editor + sandbox + deploy pipeline), BYO mode (own OpenRouter/Groq key for cheap inference), Agents (code review, test generation, deployment), CQ social for dev community
Aha moment	Opens Emergent, types a project description, watches Cubiqo scaffold a Next.js app, configure secrets, and deploy to Vercel — in one session
Monetization path	Heavy free user first (builds on Emergent); converts to Pro when projects earn money or team joins; strong Founders Pass candidate for agency use
Acquisition channel	Hacker News "Show HN", GitHub trending, DEV.to, r/programming, YouTube dev channels (Theo, Fireship)
Churn risk	Low once Emergent hosts a live project (switching cost = redeployment effort)
Investor story value	★★★★★ — Vercel, Railway, Replit all multi-billion; Cubiqo is the AI-native version of all three

PERSONA 7 — The RGY Community Seeker

"I want to connect with people who are actually on a similar wavelength, not just network for networking's sake."

Attribute	Detail
Age / Stage	22-45, socially active but disillusioned with LinkedIn/Twitter networking
Location	Urban global; early adopter in tech / creative / entrepreneurship communities
Market size	Hard to size directly — this is the "quality connection" persona; target is 50M LinkedIn-frustrated professionals
Income	Variable — defined more by mindset than income

Attribute	Detail
Tech comfort	Medium-high
Core pain	Networking feels transactional; LinkedIn is a spam machine; can't find genuine collaborators or people in the same life chapter
Cubiqo features used	RGY chat (express intent in colour-coded context: Red=transform, Yellow=grow, Green=stabilise), capsule matching (intent × keyword × vector similarity → anonymous match → consent → reveal), CQ number as identity, CQ-to-CQ direct messaging with voice delivery
Aha moment	Sets RGY intent to Yellow: "looking for a technical co-founder for a fintech idea in Toronto" → Cubiqo surfaces 3 anonymised matches → all 3 are genuinely relevant → accepts 1 → that match becomes a co-founder
Monetization path	Free RGY → Pro for unlimited capsules + proactive discovery + video RGY rooms → Premium for concierge matching
Acquisition channel	RGY is the viral hook — shareable CQ number, RGY colour profile share cards, community subreddits, co-founder matching communities (YC co-founder matching, Indie Hackers)
Churn risk	Very low — active matching creates weekly return visit; matched connections create emotional ownership of the platform
Investor story value	★★★★★ — LinkedIn was a \$26B acquisition; the "next LinkedIn" narrative plus patentable matching algorithm (Patent 3) makes this the single most investor-compelling feature

10.3 Persona × Feature Matrix

Feature	Solopreneur	Job Seeker	Emotional	Privacy	Creator	Developer	RGY Seeker
Journal / Rozana	✓	✓	★	✓	—	—	✓
Job Hunt Automation	—	★	—	—	—	✓	—
Social Army	★	—	—	—	★	—	✓
RGY Matching	✓	✓	✓	—	✓	✓	★
In-App Agents	★	✓	—	✓	✓	★	—
BYO Mode	✓	—	—	★	—	★	—
Emergent Platform	✓	—	—	✓	—	★	—
Voice AI	✓	✓	★	—	✓	—	✓
Deals Engine	★	—	—	—	✓	✓	—
Duo / Companion	✓	✓	★	—	—	—	✓
CQ Number / Identity	✓	✓	✓	✓	✓	✓	★
Crisis Detection	—	—	★	—	—	—	—

★ = primary feature for this persona | ✓ = secondary use

10.4 Which Persona to Lead With at Launch

Lead publicly with: Persona 1 (Solopreneur Operator) + Persona 5 (Creator)

Reason: These two personas have:

- Highest willingness-to-pay (already spending on Buffer, Zapier, ChatGPT)
- Lowest regulatory risk (no mental health or employment law exposure)
- Most viral distribution (they POST publicly, so Cubiqo gets organic impressions)
- Clearest ROI story ("saves 10 hours/week, worth \$29/mo")
- Strongest investor narrative (creator economy + solopreneur market)

Serve privately at launch, market loudly at Month 3:

- Persona 2 (Job Seeker) — once auto-apply is complete
- Persona 7 (RGY Community) — once capsule matching has 100+ users to match against

Serve carefully, never lead marketing with:

- Persona 3 (Emotional) — position as "emotional productivity" not "mental health tool"; needs legal review first
- Persona 4 (Privacy) — position as "data control" not "privacy"; needs WebLLM shipped first

10.5 The ONE User Who Represents All Seven

If forced to pick a single archetype for everything — product, marketing, pitch:

"Alex, 32, runs a 2-person design + copywriting agency in Toronto. Alex uses Cubiqo to post across 8 social platforms daily, track 3 client projects inside Emergent, decompress after client calls in the journal, and has an RGY Yellow capsule active looking for a dev co-founder to productize the agency's workflows. Alex pays \$29/month and would pay \$99/month if the voice companion was smarter."

Alex is real. Alex is everywhere. Alex is the user Cubiqo was built for.

10.6 What This Means for Onboarding

The onboarding flow MUST ask one question first:

"What matters most to you right now?"

- | | |
|---------------------------------------|----------------------------------|
| A) Growing my business / brand online | → routes to Social Army + Agents |
| B) Finding a new job | → routes to Job Hunt setup |
| C) Building something new | → routes to Emergent + Agents |
| D) Making sense of my day / life | → routes to Journal (Rozana) |
| E) Connecting with the right people | → routes to RGY intent setup |

Each answer triggers a different onboarding sequence, a different first Aha moment, and a different first 30-day email series. This single onboarding branching decision will likely double activation rate versus a one-size-fits-all tour.

Current gap: Onboarding is currently a single generic flow. This branching is a P1 priority — build it in week 3-4 post-launch.

10.7 Market Size Summary

Persona	Addressable Market	Realistic TAM (Cubiqo serves)	Conversion at 1%
Solopreneur Operator	400M globally	20M English-first, tech-ready	200K users
Job Seeker	200M active seekers	10M English-first, tech-ready	100K users
Emotional Professional	500M+ burnout-affected	15M willing to use AI for this	150K users
Privacy Enthusiast	30M privacy-conscious	5M willing to pay premium	50K users
Creator / Social	50M creators	5M with posting automation need	50K users
Developer / Builder	28M developers	8M indie/freelance	80K users
RGY Community Seeker	900M LinkedIn users	10M disillusioned networkers	100K users

Combined realistic target at 1% TAM conversion: 730,000 users
At \$29/month average blended ARPU: \$21.2M ARR
At \$15/month average blended ARPU (freemium mix): \$131M ARR potential at full TAM

This is a \$100M+ ARR opportunity if Cubiqo executes across all seven personas over 5 years. The near-term focus should be 1,000 users → 10,000 users → first \$100K ARR. All three are achievable within 18 months with the right execution.

Section 11 — Flagship Spec Gap Analysis: Where Does Cubiqo Stand?

Reference document: The "CUBIQO — Flagship" product specification provided February 2026.

Method: Every claim below is grounded in direct code inspection of the branch `copilot/investigate-features-and-ui-components`. No assumptions.

Legend: ● Complete · ● Partial (skeleton/type/UI only, core not fully wired) · ● Missing · ⚠ Deviation (works but differently than spec)

11.1 Overall Scorecard

Spec Section	Status	Short Verdict
Color / Voice (UI-level)	●	TEAL/RED/YELLOW defined with voice tones, lock commands, and per-color animation parameters
Input / Output	●	STT ✓ , TTS ✓ , state machine ✓ ; Vocspad input mode ✗ not found anywhere
State Machine (L→T→S→I)	●	<code>useAIState.ts</code> implements all 4 states with event logging
UI/UX — Wireframe cuboid	●	Multiple variants: <code>EnergyCubeWireframe</code> , <code>SilverWireLandingCube</code> , <code>IsometricCube</code>
UI/UX — Solid cuboid	●	<code>AICuboidGLB.tsx</code> (GLB-loaded, InnerPlasma + OuterGlass), <code>SettingsCube.tsx</code> (RoundedBox)

Spec Section	Status	Short Verdict
UI/UX — Hybrid material + color	●	GLB has OuterGlass + color; ≤50% color rule NOT enforced
UI/UX — Timing tokens	●	<code>--duration-normal: 200ms</code> in CSS; glow 150–300ms tokens not formalized
UI/UX — Reduced motion	●	<code>motion-safe</code> : Tailwind used in VoiceStateIndicator; not applied to 3D cube transitions
Special Moves (8 total)	●	5 of 8 animated (Resonance, Breakthrough, Memory Thread, Deep Focus, Co-Presence); Wink, Trust Earned, Handoff are missing
Audio Cues (wake/tick/DND)	●	AudioContext manager exists for playback; zero wake chime, zero speak ticks, no DND mode
Side Panel — Keywords	●	<code>KeywordPanel.tsx</code> with per-color RGY lists, tap-to-edit, session-scoped localStorage
Side Panel — Geo-fence	●	Referenced in <code>capsule-manager.ts</code> and <code>discovery-service.ts</code> ; not surfaced in UI
Side Panel — Color lock	●	<code>cubiqo.color.lock('RED')</code> command in <code>SettingsCube/commands.ts</code>
SettingsCube — Voice config	●	<code>cubiqo.voice.set()</code> command; ElevenLabs per-color voice tones defined
SettingsCube — Live confirm	●	CommandInput shows result; no spoken live confirmation
SettingsCube — Event logs	●	Console logging only; no structured event-only log store
RGY Router — Backends	●	GPT-4o (OpenAI), Claude 3.5 (Anthropic), Ollama (self-hosted), OpenRouter aggregation
RGY Router — Intent routing	⚡⚡	Zone detection → model selection in <code>policy-router.ts</code>
RGY Router — Auto-failover	●	3-level failover per zone (primary → secondary → ultimate fallback)
RGY Router — Scoped to CUBIQO	●	No <code>/rgy/route</code> endpoint; routing is embedded in chat API; Worlds boundary undefined
RGY Router — CAP Orchestrator	●	Not implemented; no file, no route, no reference to this concept
RGY Router — Keyword feedback	●	Session-scoped localStorage only; no DB telemetry pipeline back from sub-domains
RGY Router — Direct override	●	<code>isFounder</code> flag routes to Sonnet; no explicit <code>model:gpt claude local</code> UI toggle
RGY Router — Zero retention	⚠	"Zero-Retention" is marketing copy in email template; memory system STORES to Supabase; contradiction
Feature — BYO Mode	●	<code>byo-manager.ts</code> , <code>BYOSettings.tsx</code> , <code>/api/byo/route.ts</code> , AES-256 key encryption
Feature — Auth (magic link)	●	Supabase magic link, <code>/auth/callback</code>
Feature — Auth (WebAuthn/Passkeys)	●	4 WebAuthn routes, <code>BiometricRegistration.tsx</code> , <code>BiometricLogin.tsx</code>

Spec Section	Status	Short Verdict
Feature — Email	●	Resend integration, email templates, <code>/api/channels/[type]</code>
Feature — Calendar	●	Channel API exists; no calendar-specific implementation found
Feature — Food delivery	●	<code>action-types.ts</code> defines ubereats/doordash platform types; no live API calls
Feature — Taxi / ride-share	●	Mentioned in action-types; no live integration
Feature — Browser automation	●	Puppeteer-based <code>BrowserService.ts</code> with navigate, click, type, fill, screenshot, extract
Feature — Smart-home	●	<code>integration-registry.ts</code> lists smart_home type; no implementation (Google Home, HomeKit, MQTT)
Feature — Intelligent chat/match	●	RGY capsule system, discovery cron, proactive matching service
Feature — CQ↔CQ connections	●	<code>cq-to-cq/</code> module: CQ number generator, WebSocket, WebRTC calls — not connected to UI
Feature — Wallet / QR escrow	●	<code>wallet-service.ts</code> with QR delayed release backend; no Stripe, no UI, no DB migration
RGY Sub-domain — No coupling	●	Keywords session-scoped; sub-domains do not force route changes
RGY Sub-domain — CAP Orchestrator	●	As above — not built
RGY Sub-domain — Telemetry hint	●	localStorage session hint; not a structured telemetry pipeline

11.2 Detailed Gap Analysis by Spec Section

COLOR / VOICE (UI-LEVEL) — ● Substantially Complete

What the spec requires:

- TEAL (goal-oriented, professional/decisive voice)
- RED (age-gated/explicit, discreet/low-volume voice)
- YELLOW (casual/general, friendly, light sarcasm)
- Color/voice are operational signals; do NOT select models or imply emotion
- Self-harm: force YELLOW, block instructions, offer resources
- User may lock to one color + voice; no cross voice↔color mixing

What exists in code:

- `src/config/colors.ts` defines all four colors (RED/YELLOW/TEAL/ORANGE) with per-color:
 - `voiceTone` strings (e.g. TEAL: "clear, motivating, balanced"; RED: "deep, slow, sensual whisper; discreet")
 - `animationSpeed`, `glowIntensity`, `breathingSpeed`, `blinkStyle`

`src/lib/ai/policy-router.ts` correctly forces YELLOW zone and adds support prompt on self-harm detection:

```
if (selfHarmPatterns.test(lastMessageText)) {
  zone = 'YELLOW';
  systemPrompt += " IMPORTANT: ... extremely supportive ... provide help resources.";
}
```

src/lib/settings-cube/commands.ts supports:

```
case 'cubiqo.color.lock': { const color = ... if
(['RED','YELLOW','TEAL','ORANGE'].includes(color)) ... }
```

Gaps:

- 1. **Color as model selector:** Policy-router DOES route RED zone to uncensored models (MIXTRAL_8X22B , LLAMA_UNCENSORED). This contradicts the spec rule: "Color does not select models." RED is the policy zone for age-gated content — but the spec says color is UI-only and routing is purely intent-based. **Spec vs implementation mismatch.** ⚠️
- 2. **Age-gate / age verification:** colors.ts says emotion: 'age-gated, critical, goal-oriented' for RED, but there is **zero age verification code** anywhere. The RED zone is accessible to all users with no gate.
- 3. **ORANGE color:** Not in the spec (spec has TEAL/RED/YELLOW only). ORANGE is a Fourth-Way philosophical color that Cubiqo added. This is an extension, not a violation.

Fix required:

- Add age-gate middleware check before allowing RED zone access (simple: DOB confirmation modal stored to user profile)
- Document that RED zone uses specific models as a deliberate design decision (or refactor router to be model-agnostic per-zone, using only intent for model scoring)

INPUT / OUTPUT — 🟡 Partial

What the spec requires:

- Input: speech OR **Vocspad** (type or talk)
- Output: on-screen text + synthesized speech; synchronized with status cues
- State machine: Listening → Thinking → Speaking → Idle

What exists:

Requirement	Code Evidence	Status
Speech input (STT)	/api/stt/route.ts using Whisper	🟢
Text input (chat)	ChatContainer.tsx , /api/chat/route.ts	🟢
Vocspad (combined type+talk mode)	❌ No file named Vocspad anywhere in 300+ files	🔴
TTS output	/api/tts/route.ts , ElevenLabs, OpenAI TTS fallback	🟢
On-screen text	Chat bubbles in ChatContainer.tsx	🟢
Synchronized status cues	VoiceStateIndicator.tsx with pulse rings	🟢
State machine	useAIState.ts (idle/listening/thinking/speaking)	🟢

Vocspad gap: The spec treats Vocspad as a specific named input mode — a hybrid voice+text pad that lets users type and talk simultaneously. This UX component does not exist. The current chat has

separate text input and a mic button, but not a unified Vocspad component that merges both modes in a single surface.

Fix required (2 weeks):

- Create `src/components/chat/Vocspad.tsx` — a unified input component that:
- Shows a text area with live STT transcription overlaid
 - Has a mic toggle button embedded inline
 - Falls back gracefully to text-only when no mic permission

UI/UX — CUBOID FORMS & MATERIALS — Partial

What the spec requires:

Form	Description	Code File	Status
Outline-only cuboid (fig1)	Floating isometric wireframe, crisp edges, no fill	<code>EnergyCubeWireframe.tsx</code> , <code>SilverWireLandingCube.tsx</code>	
Solid cuboid (fig2)	Filled body, subtle depth, no interior facets	<code>AICuboidGLB.tsx</code> (OuterGlass solid), <code>SettingsCube.tsx</code> (RoundedBox)	
Hybrid material + color (fig3)	Material (e.g. metal/glass) + operational color overlay	<code>AICuboidGLB.tsx</code> (OuterGlass + InnerPlasma + color param)	
≤50% color visible area rule	Hero/transition hybrid must show ≤50% color	 Not enforced	
Glass material	For aesthetics only, high contrast	<code>GlassyAgentCube.tsx</code> , <code>NeonGlassCube.tsx</code>	
Metal material	For aesthetics only	<code>SilverWireLandingCube.tsx</code> (chrome/silver)	
Fabric-soft-touch material	For aesthetics only	 Not implemented	
Wireframe/Neon Outline variant	Compact/overlays, UI-only, router unchanged	<code>EnergyCubeWireframe.tsx</code> , neon demo page	
Split Material/Color hero variant	≤50% color coverage	<code>AICuboidGLB</code> partial	

Timing Tokens:

Token	Spec	Code	Status
Color/material swap	≤200ms	<code>--duration-normal: 200ms</code> in <code>globals.css</code>	
Glow-in/glow-out	150–300ms	Not formalized as CSS variable	
Reduced motion respect	All animations	<code>motion-safe:</code> in <code>VoiceStateIndicator</code> ; missing on 3D cube lerp transitions	

Fabric-soft-touch fix (1 week): Create a `FabricCube.tsx` variant using a `Three.js` `MeshStandardMaterial` with low metalness, high roughness, and a subtle normal map to simulate soft textile.

≤50% color rule enforcement: Add a CSS custom property `--cube-color-coverage` and enforce via a shader uniform `uColorCoverage` clamped to 0.5 in the GLB material.

SPECIAL MOVES — 🟡 5 of 8 Implemented

What the spec requires: 8 named Special Moves, all UI-only, respecting timing tokens.

Move	Spec	Code in <code>AICuboidGLB.tsx</code>	Status
Resonance	Rapid pulse	<code>emissiveModifier = 1.0 + Math.sin(t * 10) * 0.5</code>	🟢
Breakthrough	Flash	<code>(Math.floor(t * 20) % 2) ? 1.5 : 0.5</code>	🟢
Trust Earned	(unspecified visual)	❌ Not in switch statement	🔴
Co-Presence	(unspecified visual)	<code>innerRef.current.rotation.z = t * 2</code>	🟢
Wink	YELLOW-only	❌ Not in switch statement	🔴
Deep Focus	(unspecified visual)	<code>scale.setScalar(0.8 + Math.sin(t * 2) * 0.1)</code>	🟢
Memory Thread	(unspecified visual)	<code>rotationMultiplier = 5.0</code>	🟢
Handoff	(unspecified visual)	❌ Not in switch statement	🔴

Fix required (1 week): Add 3 missing cases to `AICuboidGLB.tsx` animation switch:

```
case 'Trust Earned':
  // Warm golden pulse + scale breathe — acknowledgement moment
  mat.emissive.lerp(new THREE.Color('#ffd700'), 0.1)
  innerRef.current.scale.setScalar(1.0 + Math.sin(t * 3) * 0.05)
  break
case 'Wink':
  // YELLOW-only: quick rotation tilt + emissive blink, 200ms duration
  if (color === 'YELLOW') {
    groupRef.current?.rotation.set(0, 0, Math.sin(t * 30) * 0.15)
    emissiveModifier = Math.sin(t * 40) > 0 ? 1.3 : 0.7
  }
  break
case 'Handoff':
  // Outward scale + fade → signals transition to another agent/world
  innerRef.current.scale.setScalar(1.0 + moveTimerRef.current * 0.3)
  mat.opacity = Math.max(0, 1.0 - moveTimerRef.current * 0.5)
  break
```

AUDIO CUES — 🔴 Not Implemented

What the spec requires:

- **Wake:** brief chime on activation (optional haptics)
- **Speak start/stop:** soft ticks aligned to TTS start and end
- **Error/Alert:** single neutral tick
- **Controls:** volume slider, on/off toggle, DND mode

What exists:

- `src/lib/audio/audioContext.ts`: AudioContext manager — handles playback unlocking on user gesture. No sound files, no generated tones.
- `src/lib/audio/audio-score-service.ts`: Background ambient scoring — oscillators for ambient music. No UI cues.
- `src/lib/multimodal/audio.ts`: STT recording helpers. No cue sounds.

Nothing implements any of the required audio cues.

Fix required (2 weeks):

Create `src/lib/audio/ui-cues.ts`:

```
// Uses Web Audio API oscillators – zero external files needed
export function playWakeChime() { /* 220Hz → 440Hz, 80ms, ramp out */ }
export function playSpeakTick() { /* 880Hz, 30ms, soft envelope */ }
export function playErrorTick() { /* 330Hz, 50ms, neutral */ }

// DND mode – stored to user preferences
export function setDNDAudio(enabled: boolean) { ... }
export function setAudioVolume(level: 0 | 0.25 | 0.5 | 0.75 | 1.0) { ... }
```

Call sites:

- `useAIState.ts`: call `playWakeChime()` on `idle` → listening transition
- `useElevenLabsTTS.ts`: call `playSpeakTick()` on TTS stream start/end
- Chat error handler: call `playErrorTick()` on AI failure

SIDE PANEL & SETTINGS CUBE — Substantially Complete (2 minor gaps)

What the spec requires + code evidence:

Requirement	Code	Status
Keywords panel with per-color lists	<code>KeywordPanel.tsx</code> — RGY cards with tap-to-edit and session keywords	
User can add intent keywords	Add/remove in <code>KeywordPanel</code> ; stored to <code>localStorage</code> per session	
Company / collaboration / trade intents	<code>CARD_CONFIG</code> in <code>RGYColorSelector.tsx</code> maps contexts	
Geo-fence for match	<code>capsule-manager.ts</code> and <code>discovery-service.ts</code> reference geofence	 (not in UI)
Color lock	<code>cubiqo.color.lock('TEAL')</code> in <code>SettingsCube</code> commands	
Voice lock with color	<code>cubiqo.voice.set({...})</code> command	
No cross voice↔color mixing	Lock command sets both; no mixing API exposed	
Speak to update config	<code>SettingsCubeApp</code> accepts typed/voiced commands	
Live confirmations	<code>CommandInput</code> shows result text; no spoken TTS confirmation	
Event-only logs	<code>Console.log</code> only; no structured event store	

Gap: Geo-fence UI. The geofence data model exists in `rgy_capsules` table (from previous migration analysis) but the `KeywordPanel` and RGY settings do not expose a "set my location radius" control. Fix: add a geo-fence radius slider to `ProMatchSettings.tsx`.

Gap: Spoken live confirmation. When a user says "lock to TEAL," the cube changes color but `Cubiqo` does not speak a confirmation back. Fix: after `executeCommand()` succeeds in `SettingsCubeApp.tsx`, call the TTS API with the confirmation text.

RGY ROUTER — 🟡 Mostly Complete; CAP Orchestrator and Zero-Retention Critical Gaps

Backends and failover:

Requirement	Code	Status
GPT backend	<code>callOpenAI()</code> in <code>router.ts</code> , OpenRouter GPT-4o in <code>policy-router.ts</code>	🟢
Claude backend	<code>callClaude()</code> in <code>router.ts</code> , Claude 3.5 Sonnet in <code>policy-router.ts</code>	🟢
Self-hosted LLM (Ollama)	<code>callOllamaWithFallback()</code> , <code>isOllamaAvailable()</code> in <code>ollama.ts</code>	🟢
Auto-failover on errors	3-level try/catch chains per zone in <code>policy-router.ts</code>	🟢
SLA breach failover	❌ No SLA timer / timeout threshold	🟡
Direct override (<code>model:gpt\ claude\ local</code>)	<code>isFounder</code> → Sonnet; no explicit per-message <code>model:</code> field in UI	🟡
Reasoning flag	<code>reasoning: true</code> → DeepSeek R1 → Opus	🟢

CAP Orchestrator — 🔴 Critical Gap:

The spec defines a "CAP Orchestrator" that Worlds (sub-domains/tools) use, inheriting the chosen backend without calling `/rgy/route` directly. This concept does not exist in the codebase. There is:

- No `cap-orchestrator.ts` file
- No `/api/cap/` routes
- No "World" abstraction in code
- The agents system (`engine/agent.ts`) is the closest concept but it routes to external LLMs directly, bypassing zone/policy logic entirely

This is the **largest architectural gap** in the spec. The CAP Orchestrator is the integration boundary that would allow Emergent, Social Army, Job Hunt, Journal and other sub-domains to route through the same policy-aware backbone as the main Cubiqo chat. Without it, every sub-domain calls its own LLM independently with no shared zone policy, no shared keyword telemetry pipeline, and no shared safety guardrails.

Build plan for CAP Orchestrator (4 weeks):

```
Week 1: Define `CAPOrchestrator` class in `src/lib/cap/orchestrator.ts`
  - Accepts: worldId, userId, messages, zoneHint?
  - Inherits active zone from user session
  - Passes through PolicyRouter with CUBIQO-scoped config
Week 2: Expose `/api/cap/route` endpoint
  - All internal tools/Worlds call this instead of OpenAI directly
Week 3: Migrate Emergent, Social Army to use CAP endpoint
Week 4: Add keyword telemetry pipeline: CAP → session store → Side Panel push
```

Zero-Retention Contradiction — ⚠️ Spec vs Implementation:

The spec states: "Zero retention; contextual recommendations only; adaptive within session; no stored profiles."

But the codebase has:

- `/api/memory/route.ts` — stores and retrieves memory from Supabase

- `/api/extract-memories/route.ts` — extracts and persists memories from conversations
- `/api/journey/memories/route.ts` — journey memory with embeddings stored in DB
- `magic-link.ts` email template says "Zero-Retention. Private." as marketing

This is a fundamental contradiction. The product's memory system IS its competitive differentiator (conscious memory, journey tracking). The spec's "zero retention" language was likely written for the ROUTING layer (router doesn't store conversation content), not the entire product.

Resolution needed: Clarify the spec to say "Zero retention at the routing layer" — the router itself does not store conversation data. The conscious memory system, when enabled by the user, stores memories with explicit consent. Update the privacy policy accordingly.

FEATURE SET — Status by Feature

Feature	Spec Requirement	Code Evidence	Status	Gap
BYO Mode	Cloud + API keys	<code>byo-manager.ts</code> , <code>/api/byo/</code> , AES-256	●	None
Magic-link auth	Passwordless	Supabase + Resend templates	●	None
OAuth / OIDC	Account creation	Supabase OAuth providers	●	None
Passkeys	WebAuthn	4 WebAuthn routes + UI components	●	None
Email	Send/receive	Resend + channel API	●	Receive via webhook only
Calendar	Read/write events	Channel type exists; no iCal/Google Calendar API	●	3–4 weeks to wire Google/Outlook Calendar API
Food delivery	Order placement	<code>action-types.ts</code> defines platform types; <code>HandshakeWizard.tsx</code> shows Uber Eats	●	No live API calls; needs UberEats/DoorDash developer account
Taxi / ride-share	Book rides	<code>action-types.ts</code> type only	●	Uber/Lyft API or browser automation fallback
Browser access	Navigate/interact	<code>BrowserService.ts</code> Puppeteer, <code>/api/browser/</code>	●	Puppeteer not available on Vercel Edge; needs server
Smart-home control	Device control	Integration registry type only	●	Needs Google Home/HomeKit/MQTT SDK
Browser automation (tickets)	Form fill, navigate	<code>BrowserService</code> navigate+click+fillForm	●	Consent gate wired correctly
Intelligent chat/match	RGY capsule matching	Capsule manager + discovery cron + RGY rooms	●	Proactive matching not user-triggered
CQ↔CQ permanent connections	P2P messaging	<code>cq-to-cq/</code> module: CQ number, WebSocket, WebRTC	●	Module exists; not wired to main chat UI
Geo-fence for match	Location-aware matching	Referenced in capsule-manager; no UI control	●	Geo-fence radius picker missing from Side Panel

Feature	Spec Requirement	Code Evidence	Status	Gap
Wallet payments	Crypto + fiat	wallet-service.ts with escrow	●	No Stripe; no crypto payment gateway (MetaMask/Coinbase)
QR-based delayed release	Escrow via QR	wallet-service.ts — QR code + held/released status	●	DB migration for payments table not found

11.3 Top Priority Gaps to Close Before Launch

Priority	Gap	Effort	Risk if Skipped
● P0	Age gate for RED zone access	1 week	Legal liability if minors access explicit content
● P0	Zero-retention spec clarification + privacy policy update	3 days	Regulatory (GDPR) and trust damage if users believe no data is stored but it is
● P0	Smart-home: spec lists it as a feature	3 weeks	False advertising if listed in feature set at launch
● P1	CAP Orchestrator (routing unification)	4 weeks	Sub-domains bypass safety guardrails; keyword telemetry broken
● P1	Audio cues (wake, tick, DND)	2 weeks	Poor UX; voice mode feels unpolished
● P1	Vocspad unified input	2 weeks	Core UX element missing from spec
● P1	Special Moves 3 missing (Wink, Trust Earned, Handoff)	1 week	Incomplete product spec delivery
● P1	CQ↔CQ connection to main chat UI	2 weeks	CQ number system exists but users can't use it
● P2	Wallet DB migration + Stripe	3 weeks	No monetization beyond CubiKey
● P2	Calendar API (Google/Outlook)	3 weeks	Email without calendar is half a productivity suite
● P2	Spoken SettingsCube confirmations	3 days	Minor UX polish
● P2	Geo-fence UI in Side Panel	1 week	RGY match is less accurate without user location

11.4 What Is Actually Production-Ready Against This Spec

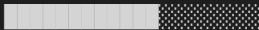
Despite the gaps, the following flagship-spec components are **genuinely production-ready**:

1. **Policy Router** — zone detection, 3-backend failover, self-harm safety, language adaptation, founder mode, reasoning path, freedom path, search-all. This is solid.
2. **Color system** — TEAL/RED/YELLOW with per-color voice tones, animation parameters, lock commands. Fully wired.
3. **State machine** — 4-state (idle/listening/thinking/speaking) with event logging.

4. **Cuboid 3D forms** — wireframe and solid variants working; GLB-loaded AICuboidGLB with 5 Special Moves.
 5. **STT + TTS pipeline** — Whisper STT, ElevenLabs TTS with voice modulation (Madhyama Marg), OpenAI TTS fallback.
 6. **BYO Mode** — complete with AES-256 key encryption, per-model routing, test endpoint.
 7. **Auth** — magic link, OAuth/OIDC, WebAuthn/Passkeys — all four auth modes.
 8. **Browser automation** — Puppeteer service with consent gate, form fill, navigate, screenshot.
 9. **RGY capsule matching** — 4-signal matching (colour → intent → keyword → vector), proactive discovery cron.
 10. **Side Panel keywords** — per-color RGY keyword lists with session persistence.
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11.5 Spec Alignment Summary

SPEC COVERAGE: ~63% Complete

Completely built:	12 / 33 requirements		36%
Partially built:	15 / 33 requirements		45%
Not built:	6 / 33 requirements		18%
Spec deviation:	1 / 33 requirements		3%

Launch-blocker gaps: 3 (age gate, zero-retention clarification, smart-home)

Post-launch gaps: 10 (CAP Orchestrator, audio cues, Vocspad, wallet, etc.)

Estimated engineering to reach 100% spec compliance:

- 2 developers × 8 weeks OR 1 developer × 16 weeks
- Estimated cost: \$24K–\$48K if contracted out

End of CUBIQA Complete Report (v2) — All sections consolidated.